NeuroVLM: A generative vision-language framework for human neuroimaging

Point-by-point response to the reviewers

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Letter to the editor

We thank the editor and reviewers for their thoughtful comments. The revisions and response address every comment. The main changes are:

- Independent voxel-space evaluation of text-to-brain maps against simple baselines and existing, published models (1.1).
- Retrieval measured by recall@k, mean reciprocal rank, and median rank, also against hard negatives (1.1).
- A comparison of generated and retrieved captions (3.3).
- A data-flow diagram of every training stage (1.2).
- NeuroVault results reported as an upper bound, a lower bound, and a multi-target evaluation, with map types handled separately (1.3).
- A leakage audit beyond DOI matching (2.4).
- Out-of-distribution and failure-case analyses (1.4 and 2.12).
- Corrected numbers, names, and claims throughout.

We also updated the model and its evaluation:

- We added a latent-space loss when training the text-to-brain head, which makes its maps more accurate (1.1).
- Generated maps are calibrated to the scale of the training maps, with one factor per model fitted on the validation set.
- An MLP projector replaces the Qformer in both brain-to-text stages (2.1).
- The language model still generates every title, steered toward the canonical network and region names by masking tokens that cannot continue one (2.12).
- Decoding uses Latin-script tokens only (2.12).
- The synthetic training maps had included held-out papers. We rebuilt them from training papers only and retrained the models that use them (2.4).
- Every evaluation leaves out the duplicated papers found by the leakage audit (2.4).
- All paper figures were redrawn from the final model.

Comments are numbered in order within each reviewer, minor comments and code availability included. For example, 1.5 is the first minor comment of Reviewer 1. Each comment is quoted in a shaded box, followed by our response and the changes to the manuscript. Comment numbers link to those comments. Figures and tables in this letter are numbered R1, R2, and so on.

Line and page numbers in the changes to the manuscript refer to the revised manuscript with changes marked, which shows additions in blue and deletions in red.

Reviewer 1

Reviewer's comment

*This is an ambitious and timely manuscript introducing **NeuroVLM**, a unified framework for contrastive text–brain retrieval, text-to-brain generation, and brain-to-text generation. The integration of these capabilities within a modular architecture is appealing, and the use of a relatively large corpus of 30,826 publication-level text–neuroimage pairs is a notable strength. The public release of the code, pretrained models, and datasets should also be valuable to the community. The results provide encouraging evidence that the model learns meaningful associations between scientific text and coarse spatial brain representations. However, I think several central aspects of the evaluation and data construction require further clarification or analysis before the stronger claims regarding map generation and neuroimage interpretation are fully supported.*

Response: We added the requested analyses and clarifications (1.1 to 1.10).

1.1: More independent evaluation of the generative outputs

Reviewer's comment

1. More independent evaluation of the generative outputs is needed

The current generative evaluation is performed partly within NeuroVLM's own learned representation space. Generated maps and text are embedded using the model's contrastive projection heads and then ranked against candidate items in that same space. This is a useful measure of internal consistency, but it does not independently establish that a generated map is anatomically accurate or that a generated description faithfully reflects the source image.

The matched-versus-null results are statistically strong, although some absolute differences are relatively modest—for example, Dice is 0.47 versus 0.40, BERTScore is 0.56 versus 0.52, and semantic cosine is 0.47 versus 0.40. These findings support above-null alignment, but additional validation would help establish neuroscientific specificity.

I encourage the authors to add some combination of:

- *Evaluation using independent image and text representations not used during NeuroVLM training.*
- *Blinded expert ratings of anatomical accuracy, semantic accuracy, specificity, and unsupported claims.*
- *Hard negatives drawn from the same cognitive domain or from studies with similar spatial patterns.*
- *Comparison between generated outputs and their nearest training examples.*
- *Strong simple baselines, including nearest-neighbor retrieval, mean or domain-specific maps, linear regression from text embeddings to spatial coefficients, and atlas-based labeling.*

It would also be useful to report conventional retrieval measures such as recall@1, recall@5, recall@10, mean reciprocal rank, and median rank, in addition to the area under the full recall@k curve.

Response: We agree. Fig. R1 and Table R1 score text-to-brain maps in voxel space, against each test paper's reconstructed map. NeuroVLM leads, narrowly ahead of linear regression and ahead of Text2Brain, mean and domain maps, atlas labeling, Neurosynth, NeuroQuery, and the nearest training paper (Dice 0.53 against at most 0.49). Neurosynth, NeuroQuery, and Text2Brain are used as published, although their corpora each hold nearly half the test papers.

Independent evaluation. Both directions are also scored outside NeuroVLM's representation space. Maps are compared in voxel space, and NeuroVLM also leads against the raw coordinate maps. MiniLM cosine and BERTScore, whose encoders were not used in training, compare generated captions with the paper's text (3.3).

Matched and shuffled pairs. The submission's BERTScore null came from an earlier generation run. With the null computed from the same captions, BERTScore separates matched from shuffled pairs only weakly (0.553 against 0.548). BERTScore mostly rewards the style of the reference, an LLM summary. Even random training texts in that style score higher (Table R11).

Modest differences. The narrow gaps are real. Captions describe the map rather than the study, so they share little content with the paper's text. Unrescaled BERTScore stays near 0.58 even for two unrelated papers. The matched caption scores higher for 57% of maps by BERTScore, and more often by MiniLM. The paired test over all test maps resolves this small shift (Fig. R2).

Expert ratings. We did not run blinded expert ratings, because they are time consuming and hard to scale. Instead, we used existing expert labels and automatic checks. For anatomical and semantic accuracy, region titles are checked against anatomical atlases and network titles against expert atlas labels (3.6). Hard negatives test specificity. Unsupported claims are counted as study details, such as a sample size, that a map cannot support (3.3).

Hard negatives. Table R2 scores retrieval among all test papers and among hard negatives, the test papers of the query's own cognitive domain. Hard negatives are only slightly harder than random papers of the same number (recall@10 0.24 against 0.28).

Nearest training examples. Generated maps match each test paper's map better than the map of the nearest training paper does (Table R1). By MiniLM cosine, generated captions are as close to the paper's text as the nearest training caption. However, 57% of these retrieved captions report details of another study, against 2% of generated captions (3.3).

Simple baselines. Fig. R1 and Table R1 include each suggested baseline. The linear regression maps the SPECTER2 text embedding to the autoencoder latent, the spatial coefficients of each map. We fit it as ridge regression, which is linear regression with a small penalty against overfitting, chosen on the validation set. NeuroVLM's final head learns the same mapping through a hidden layer, with a loss that also scores the decoded map.

Retrieval measures. Table R2 reports recall@1, recall@5, recall@10, mean reciprocal rank, and median rank in both directions. Fig. R1 and Table R2 are new in the Supplementary Information.

Table R1. Text-to-brain maps on the PubMed test set. Mean Dice and Pearson r between each method's map and the test paper's reconstructed map ($n = 2{,}971$), as in Fig. R1. The nearest training paper is the one whose text is closest to the query.

Method	Dice	Pearson r
NeuroVLM	0.531	0.322
Simple baselines		
Linear regression (ridge)	0.526	0.311
Domain mean map	0.448	0.245
Atlas-based labeling	0.431	0.175
Global mean map	0.420	0.251
Nearest training paper	0.395	0.114
Published models		
Neurosynth	0.418	0.175
NeuroQuery	0.411	0.179
Text2Brain	0.485	0.293
Shuffled null	0.408	0.122

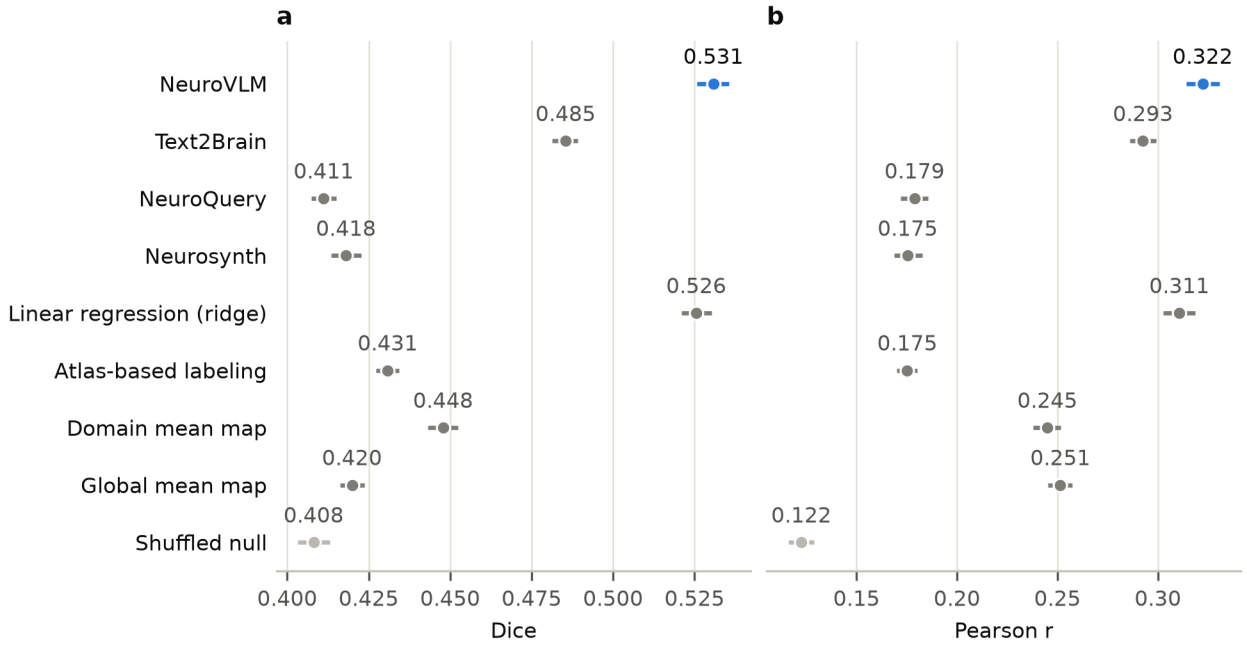


Fig. R1. Text-to-brain baselines. Text-to-brain maps from NeuroVLM and other methods, scored against each test paper's reconstructed map ($n = 2{,}971$). Panels show (a) Dice and (b) voxelwise Pearson r, with means and 95% bootstrap intervals. Dice uses each map's top 30% of voxels. Neurosynth, NeuroQuery, and Text2Brain were used as published. Their corpora include 1,386 (Neurosynth) or 1,295 (NeuroQuery and Text2Brain) test papers, which can only favor them. Text2Brain's training split is unreleased, so instead of retraining we scored the 1,157 papers in neither corpus, and NeuroVLM still beats all three.

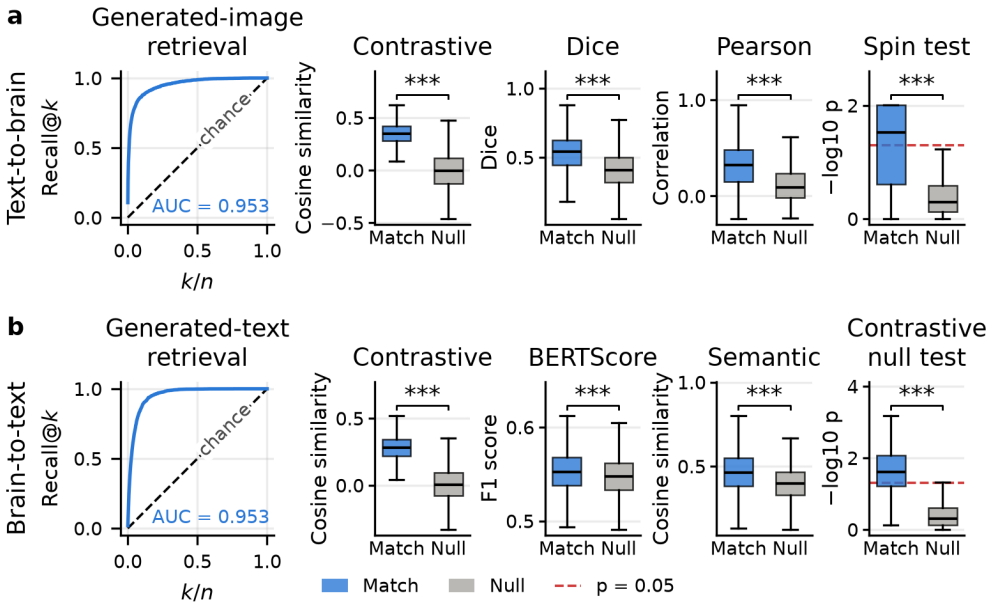


Fig. R2. New Fig. 4. Text-to-brain (a) and brain-to-text (b) generation on the PubMed test set ($n = 2{,}971$). Boxes compare matched (Match, blue) and shuffled pairs (Null, gray). Stars mark $p < 0.05$, 0.01, and 0.001.

Table R2. Contrastive retrieval on the PubMed test set. Recall@k, mean reciprocal rank (MRR), and median rank of the final model. Hard negatives are test papers from the query's own cognitive domain, and random negatives match their number.

	Recall@1	Recall@5	Recall@10	MRR	Median rank
All candidates (2,971 test papers)					
Text → brain	0.018	0.053	0.088	0.045	230
Brain → text	0.013	0.052	0.088	0.041	217
Chance	<0.001	0.002	0.003	0.003	1,486
Hard negatives (papers of the query's own domain, mean 585)					
Text → brain	0.054	0.160	0.242	0.118	44
Brain → text	0.059	0.176	0.263	0.126	40
Chance	0.003	0.017	0.034	0.019	250
Random negatives (random papers, as many as the hard negatives)					
Text → brain	0.069	0.196	0.283	0.142	35
Brain → text	0.071	0.201	0.289	0.144	35
Chance	0.003	0.017	0.034	0.019	250

Changes to the manuscript: Methods, Text-to-Brain, l. 365–385 (final head), and Performance Metrics, l. 398–404 and l. 415–426. Results, Generative Models: PubMed, l. 596–607, and Fig. 4 caption (p. 15). Supplementary Information, Projection Head, l. 1332–1339 (training of the final head), and Generative Performance, l. 1448–1450 (BERTScore null), with Table S4 (p. 32). New Supplementary Section Text-to-Brain Baselines and Retrieval, l. 1843–1859, with Fig. S6 (p. 42) and Table S8 (p. 43).

1.2: Publication-level supervision and dataset separation

Reviewer's comment

2. The publication-level supervision and separation of datasets across training stages need clarification

The primary training pairs associate publication titles and abstracts with maps reconstructed from reported coordinate tables. Because a single neuroimaging publication may include multiple experiments, contrasts, populations, and positive or negative effects, the correspondence between an abstract and one aggregated map may be substantially many-to-many.

Please clarify:

1. *Whether coordinates were combined across all experiments and contrasts within each publication.*
2. *How positive and negative effects were handled.*
3. *How multiple coordinate tables associated with one DOI were treated.*
4. *The final numbers of publications, experiments, contrasts, and coordinate tables.*
5. *Whether a contrast-level evaluation can be provided for a subset of the data.*

The complete train–test separation across model components also needs to be documented more clearly. In particular, Qwen3-0.6B was adapted using 1.2 million neuroscience articles. Please confirm that held-out PubMed and NeuroVault publications were excluded from this language-model corpus. Similarly, the manuscript should explain how test concepts and documents were excluded when constructing the synthetic fine-tuning data and canonical prototype bank. The synthetic stage uses generated maps and descriptions of common networks, regions, and cognitive concepts, creating another possible route through which test information could enter the system.

A data-flow diagram showing the datasets used for autoencoder training, contrastive training, language-model adaptation, Qformer training, synthetic fine-tuning, validation, and final testing would substantially improve transparency.

Response: 1. Experiments and contrasts. Yes, each paper's map combines the coordinates of all its experiments and contrasts. The released coordinates carry no experiment or contrast labels. Each map matches the paper-level text it is paired with.

2. Positive and negative effects. Positive and negative effects are pooled, because the released coordinates carry no sign labels. Each map therefore shows where effects lie, not their sign. The Methods state this.

3. Multiple coordinate tables. All coordinate tables of a paper are pooled into one map. Among open-access papers with a coordinate table, 42% report two or more.

4. Final counts. Table R3 gives the final counts, 30,826 publications with 857,298 peaks. Coordinate tables can be counted only in papers with open-access full text, and the table gives that count. Experiments and contrasts are unlabeled, so they cannot be counted.

5. Contrast-level evaluation. A contrast-level evaluation is possible only on NeuroVault, whose maps are released per contrast (1.3). With every map as its own target, the model beats chance in both directions (AUC 0.62 and 0.63).

Separation of datasets. Fig. R3 is the requested data-flow diagram. No validation or test paper trains the autoencoder, projection heads, or projectors. The language model's training script removes the validation, test, and NeuroVault evaluation papers from its corpus. Generated captions share only short phrases with their abstracts (median two words).

Synthetic data and prototype bank. The synthetic maps use training papers only (2.4). The prototype bank holds network, region, and function texts, not test documents. It names the networks that label the evaluation atlas maps, but no atlas network map trains any component.

Table R3. Final counts of the PubMed data. Peaks are coordinate rows, duplicates included. Tables are counted in the open-access full texts. Papers with more than 100 peaks or an empty map are in no split.

Quantity	Released data
Publications / peaks	30,826 / 857,298 (median 17 peaks per paper)
Coordinate tables	20,541 in the 11,943 open-access papers with a coordinate table (42% have two or more)
Experiments, contrasts	not labeled (only paper identifiers and coordinates are released)
Train / validation / test papers	23,894 / 2,987 / 2,987, and 958 in no split (2,971 test papers after the leakage audit)

Changes to the manuscript: Methods, Datasets, l. 137–144 (pooling per paper and splits), and Training, l. 169–181 (synthetic maps from training papers only, exclusions, and the evaluation papers removed from the language-model corpus). Supplementary Information, Coordinate Tables, l. 1044–1047. Discussion, l. 877–884. New Supplementary Section Data Flow and Training Details, l. 1745–1753, with Fig. S5 (p. 39) and Table S6 (p. 40).

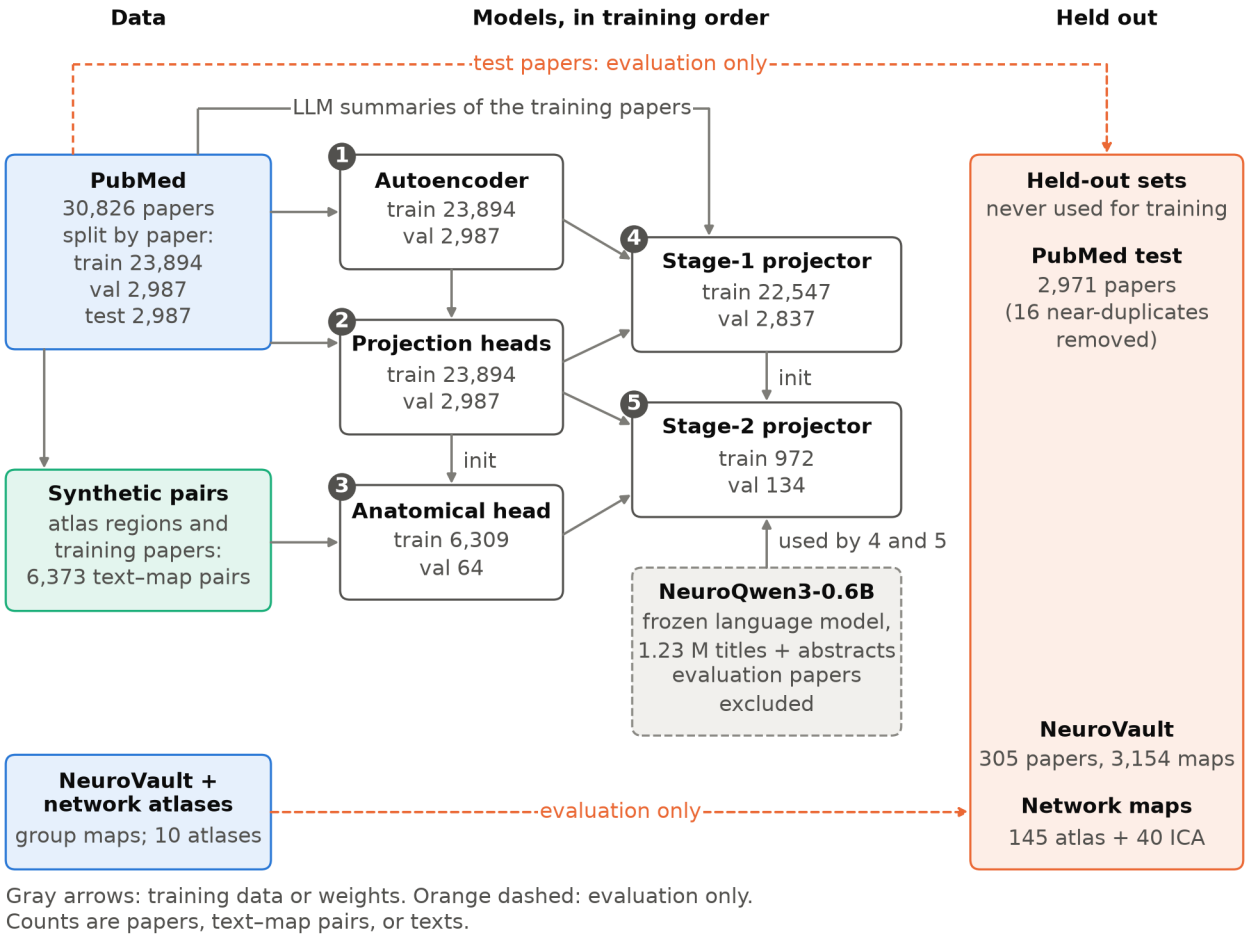


Fig. R3. Data flow of the final model. Boxes 1–5 are the trained components in training order, with their training and validation sizes. Gray arrows carry training data or weights, and orange dashed arrows mark sets used only for evaluation.

1.3: NeuroVault evaluation

Reviewer's comment

3. The NeuroVault evaluation would benefit from target-independent selection and map-type-specific preprocessing

The supplementary methods state that when several NeuroVault images were associated with the same title and abstract, the contrastive model selected the most similar image. The manuscript appropriately acknowledges that this inflates performance and provides an upper bound.

I suggest retaining this as an exploratory upper-bound analysis while adding a target-independent evaluation. For example, the authors could use contrast-specific NeuroVault metadata, evaluate all eligible images, apply a predefined image-selection rule, or use a multiple-instance retrieval metric.

The preprocessing of the NeuroVault images also requires reconsideration. The dataset contains t-, z-, F-, p-, and beta maps, as well as ROI masks and parcellations, but all maps are processed with a 99th-percentile threshold, cluster filtering, and binarization. These map types have substantially different meanings. In particular, the largest values in a conventional p-value map generally indicate weaker evidence, signed t-, z-, and beta maps may contain meaningful negative effects, and parcellations represent categorical labels rather than statistical intensity.

Please clarify whether map-specific transformations were applied before thresholding. If not, I recommend separate preprocessing and reporting for signed statistical maps, p-maps, masks, and parcellations. This would make the NeuroVault generalization results easier to interpret and ensure that spatial information is treated appropriately for each map type.

Response: We keep the model-selected map as an upper bound and add a lower bound, each paper's worst-matching map (Fig. R4). We add two target-independent scores. Multi-target retrieval counts any of a paper's maps as a hit, and the per-map score makes each map its own target. The model beats chance under every rule. Per-map text-to-brain and brain-to-text retrieval reach an area under the recall curve (AUC) of 0.62 and 0.63.

Map types. The submission thresholded every map alike, after removing masks, parcellations, anatomical maps, and variance maps. P-maps are inverted to $1 - p$ and signed maps keep their positive tail, as before. Using both tails, or adding negative tails as extra maps, reduces evaluation performance. Table R4 reports each map type separately, and Fig. 5b uses the same handling.

Maps labeled Other. Other is a type that NeuroVault uploaders choose, not a missing one. We typed these maps from their file names, descriptions, and values. Most are t, z, or F maps, effect estimates, or components (Table R4). Mean or variance images and maps we could not type are excluded, like the variance maps above. This moves every AUC by at most 0.004.

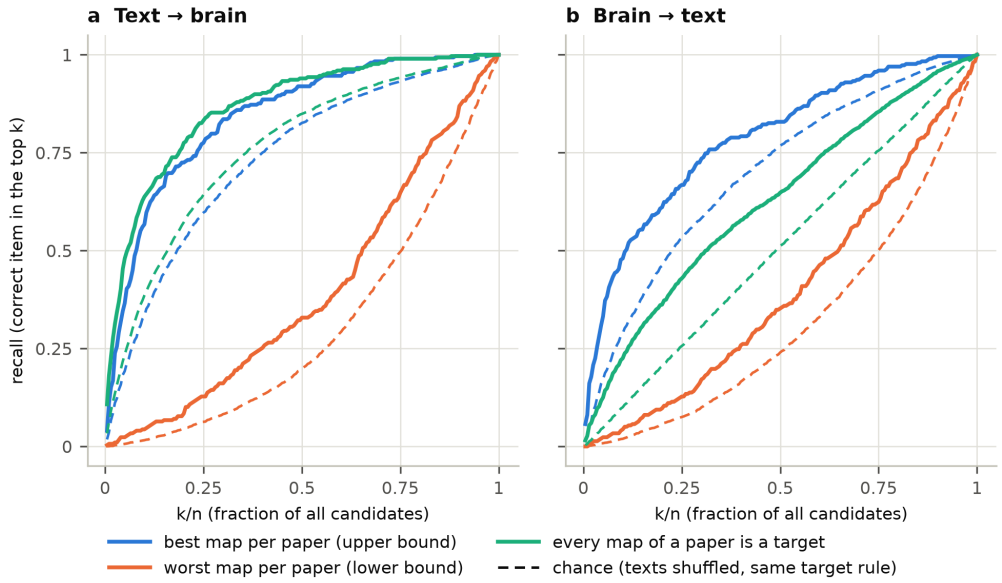


Fig. R4. NeuroVault retrieval. Recall for (a) text to brain and (b) brain to text, with each paper's best, worst, or every map as target. Dashed lines show chance from shuffled texts. The best map, or any of a paper's maps counting as a hit in (a), lifts chance above the diagonal, and the worst map lowers it.

Table R4. NeuroVault results by map type. AUC with every map as its own target (chance 0.50). Other marks maps their uploaders labeled Other, typed from file names, descriptions, and values. Masks, parcellations, anatomical and variance maps, mean images, and untyped maps were excluded.

Map type	Maps	Papers	Handling	AUC, text → brain	AUC, brain → text
T	1,761	189	positive tail	0.64	0.64
Z	811	75	positive tail	0.57	0.58
beta, univariate	43	5	positive tail	0.61	0.77
F	40	17	positive tail	0.64	0.55
beta, multivariate	14	5	positive tail	0.84	0.83
p	3	2	inverted (1 − p)	0.64	0.60
1 − p	2	1	used as is	0.54	0.44
Other, t, z, or F	204	23	positive tail	0.63	0.70
Other, effect estimate	103	15	positive tail	0.69	0.62
Other, component or gradient	93	8	positive tail	0.49	0.51

Changes to the manuscript: Fig. 5b caption (p. 16). Results, Contrastive Model: NeuroVault, l. 754–767. Supplementary Information, NeuroVault preprocessing, l. 1177–1192 (map types and exclusions) and l. 1204–1210 (bounds). New Supplementary Section NeuroVault Bounds and Map Types, l. 1954–1963, with Fig. S7 and Table S9 (p. 44).

1.4: Scope of the generative claims

Reviewer's comment

4. The scope of the generative claims should more closely reflect the available training targets

The authors appropriately acknowledge that the training images are proxy maps reconstructed from reported peak coordinates. Such maps omit subthreshold structure, cluster shape, effect size, uncertainty, contrast definitions, and other study-specific information. The additional smoothing, DiFuMo reconstruction, resampling, and binarization further emphasize broad spatial support rather than complete statistical information.

Given these limitations, the generated outputs may be more accurately described as ***coarse spatial-support predictions*** *or* ***spatial priors*** *rather than complete statistical maps. This framing would still support useful applications in literature navigation, network labeling, and exploratory hypothesis generation, while avoiding the implication that effect size, sign, uncertainty, or a study-specific statistical contrast has been reconstructed.*

Similarly, the brain-to-text outputs appear designed to provide concise, high-level network or functional descriptions rather than detailed interpretations of individual experiments. The manuscript should distinguish broad network labeling from study-specific interpretation and mechanistic inference.

The authors already note that the model can generate plausible maps for essentially any text input and should not be viewed as an oracle. A small quantitative analysis of failure cases, out-of-distribution inputs, and abstention or confidence calibration would strengthen this important caution.

Response: We agree and adopt this framing. The Abstract, Results, and Discussion call text-to-brain outputs coarse spatial priors, without effect size, sign, or uncertainty. They call brain-to-text outputs broad network-level labels, not interpretations of a specific study or mechanism. Failure cases are not rare. In 16% of test-set captions, the similarity to the paper's text is negative (2.12).

Out-of-distribution inputs. Fig. R5 flags a map when it correlates poorly with its autoencoder reconstruction or its latent lies far from the training latents. It flags a text that lies far from every training abstract. Thresholds set on the validation split flag about 5% of held-out test inputs. They

flag every shuffled, white-noise, and rescaled map, every off-topic text, and 61 to 80% of rotated or spun maps.

Abstention and confidence. Flagged inputs can be withheld, but the scores are warnings, not calibrated probabilities. Within the training distribution, the peak of a calibrated text-to-brain map works as a confidence score, since it tracks the map's retrieval rank (Spearman 0.63). It does not detect off-topic text, whose maps mostly peak within the range of test abstracts.

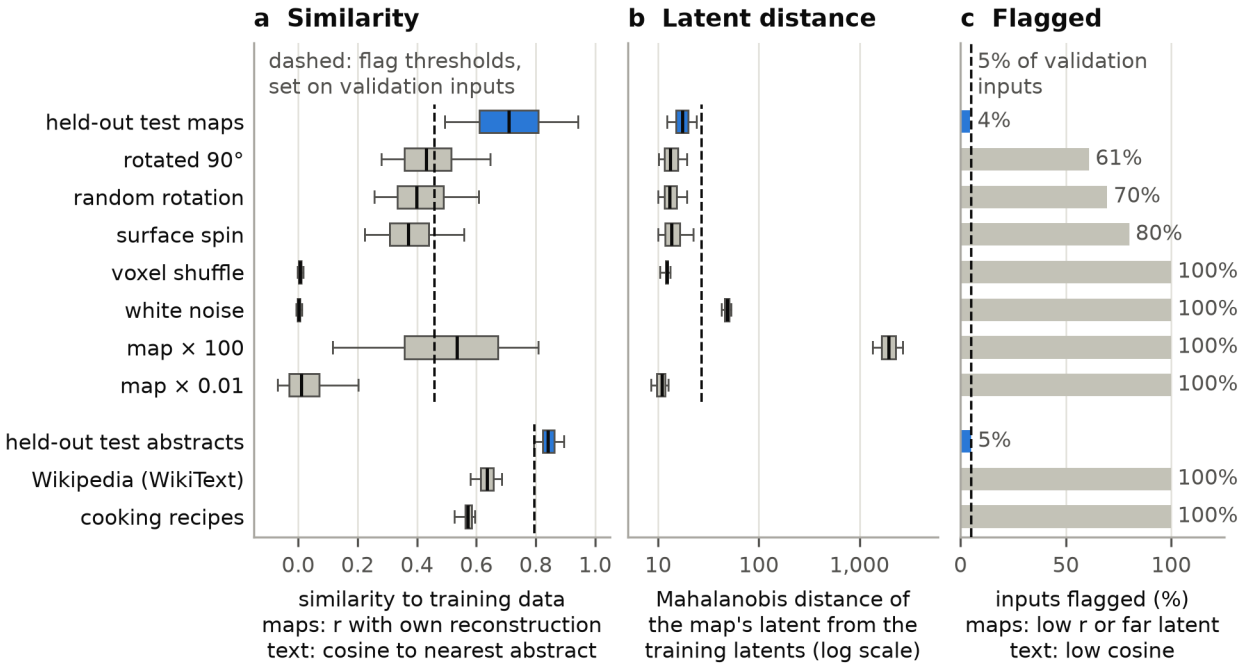


Fig. R5. Out-of-distribution warnings. Panels show (a) each input's similarity to the training data, (b) the distance of a map's latent from the training latents, and (c) the share of inputs flagged. Dashed lines mark thresholds set on the validation split. Blue marks held-out test inputs.

Changes to the manuscript: Abstract, l. 28–39. Introduction, l. 112–117. Methods, Text Decoder, l. 321–327 (out-of-distribution warnings), and Text-to-Brain, l. 374–385. Results, Generative Models, l. 432–434 (spatial priors, network-level labels). Discussion, limitations, l. 858–862 and l. 898–909. New Supplementary Section Out-of-Distribution Inputs, l. 2027–2040, with Fig. S8 (p. 45).

1.5: Inconsistent dataset and metric values

Reviewer's comment

*1. **Please reconcile inconsistent dataset and metric values.** The main methods list 3,000 NeuroVault maps and 100 network images, whereas the supplement reports 4,553 NeuroVault maps and the results describe 300 network images. The autoencoder AUROC is also reported as 0.94 in the Figure 5 caption and 0.975 in the main text.*

Response: We recomputed every count from the released data and corrected the text and captions (Table R5). The supplement's 4,553 maps predate image quality control and the removal of papers already in PubMed. Neither network-image count can be reproduced, so the released counts replace both. Both AUROCs are correct but refer to different datasets, which the Fig. 5 caption names.

Table R5. Corrected counts and AUROCs. Counts are recomputed from the released data. The AUROC is the autoencoder's voxelwise reconstruction AUROC.

In the paper	Correct value
3,000 or 4,553 NeuroVault maps	3,183 released maps from 312 papers, 3,154 maps from 305 papers after the leakage audit, and 3,074 maps from 298 papers for retrieval
100 or 300 network images	185 network maps, made of 145 atlas maps (119 with a network label) and 40 ICA components
AUROC 0.94 and 0.975	both correct, 0.94 on NeuroVault and 0.975 on the PubMed test set (0.974 with the final model)

Changes to the manuscript: Methods, Datasets, l. 132–144. Results, Contrastive Model: Network Labeling, l. 735–747, and Autoencoder, l. 826–830. Figs. 4 and 5 captions (pp. 15–16). Supplementary Information, NeuroVault preprocessing, l. 1177–1192 and l. 1209–1211.

1.6: SPECTER and SPECTER2

Reviewer's comment

*2. **Please use SPECTER and SPECTER2 consistently.** Section 2.6 describes SPECTER, whereas the text-to-brain methods and supplement describe SPECTER2.*

Response: Every text-encoding step uses SPECTER2, the allenai/specter2_aug2023refresh model with its ad hoc query adapter. The original SPECTER is never used. We corrected every mention to SPECTER2.

Changes to the manuscript: Introduction, l. 107–111. Methods, Text Encoder, l. 220–227, and Text-to-Brain, l. 365–373. Supplementary Information, Text preprocessing, l. 1085–1088. Fig. S1 caption (p. 23).

1.7: Two methodological analogies

Reviewer's comment

*3. **Two methodological analogies should be revised.** Transformer language models still depend on a fixed tokenizer vocabulary, despite their ability to represent context and novel compositions. In addition, a separately trained projection head is not technically equivalent to LoRA, although both approaches can preserve the weights of a larger pretrained encoder.*

Response: We agree with both points. We deleted the LoRA analogy and rewrote the tokenizer sentence. It states that contextual encoders also use a fixed vocabulary and that their advantage is composing words in context. SPECTER2 beats a context-free word2vec encoder (3.4). Swapping one word moves the generated map as expected, but negation has no effect (3.5).

Changes to the manuscript: Introduction, l. 81–84 (tokenizer sentence). Methods, Text-to-Brain, l. 365–385 (LoRA sentence deleted, projection head named).

1.8: Edge-device claim

Reviewer's comment

*4. **The edge-device claim would benefit from empirical support.** Please report total memory requirements, inference latency, hardware, numerical precision, and any quantization used. The relatively small text encoder is attractive, but the complete brain-to-text pipeline also includes the Qformer and Qwen language model.*

Response: We timed both full pipelines, projector and language model included, on a consumer GPU and a desktop CPU at several precisions (Table R6). We used the released weights, whose projector and text-to-brain head match the final ones in size. On the CPU, int8 quantization runs brain-to-text fastest, with the same memory as bf16. No phone or embedded device was tested, so "edge devices" becomes "a consumer CPU" in the Methods.

Table R6. Cost of one query. Median latency at batch 1 and peak memory of weights and activations, not file size. Brain-to-text decodes 128 tokens greedily. The GPU is an RTX 5090 and the CPU an 8-thread Ryzen 7 9800X3D. Dynamic int8 quantization skips NeuroQwen3's token embedding, adds an int8 copy of its tied output layer, and skips SPECTER2's adapter-wrapped layers, so it saves no memory over bf16.

Pipeline, one query	Parameters	GPU, bf16	CPU, fp32	CPU, bf16	CPU, int8
text-to-brain	171 M	7 ms, 0.4 GB	135 ms, 0.6 GB	53 ms, 0.4 GB	120 ms, 0.5 GB
brain-to-text, 128 tokens	685 M	3.2 s, 1.3 GB	13.1 s, 2.7 GB	6.3 s, 1.4 GB	3.5 s, 1.4 GB

Changes to the manuscript: Methods, Text Encoder, l. 226–230 (edge-device sentence replaced by CPU costs). Discussion, l. 935–936.

1.9: Figure text size, full descriptions, affiliations

Reviewer's comment

5. Some text in Figures 2, 3, and S4 is difficult to read at normal manuscript scale. Providing the complete generated descriptions in a supplementary table would be helpful. Please also correct the remaining "Organization" placeholders in the affiliation block.

Response: We redrew Figs. 2, 3, and S4 at the journal's text width with larger text. Figs. R6, R7, and R8 show the new Figs. 2, 3, and S4. Supplementary Table S5 gives the complete texts of Fig. 2, and Figs. 3 and S4 print each generated text in full beside its map. We also corrected the affiliations.

Changes to the manuscript: Affiliations, l. 16–17. Figs. 2, 3, and S4, redrawn (p. 11, p. 12, and p. 34). Results, Generative Models: Networks, l. 442–448. New Supplementary Table S5 with the complete texts of Fig. 2, l. 1567–1570 (pp. 35–38). Figs. 4 and 5 captions (pp. 15–16).

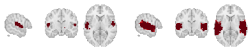
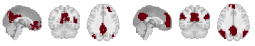
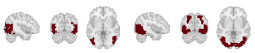
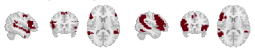
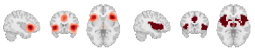
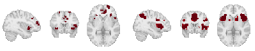
	Generated	Generated (brain-to-text)		
True	(text-to-brain)	Network	Anatomy	Cognition
Auditory		**Auditory Network.** The auditory network includes Heschl's gyrus, superior temporal gyrus, planum temporale, and inferior frontal cortex. It is commonly associated with the processing of sound, speech, and environmental sounds. The network is often described as a hierarchical system for auditory perception. Its coordinated activity is thought to support the integration of auditory information across the brain.	**Superior Temporal Gyrus.** The superior temporal gyrus is centered on the superior temporal sulcus. It is commonly associated with auditory processing, speech perception, and sound-related associations. Its anatomy supports representation of acoustic features such as frequency, timbre, and pitch.	**Early Auditory Processing.** Early auditory processing is the analysis of sound input at high frequencies. It involves primary auditory cortex in Heschl's gyrus, superior temporal gyrus, and planum temporale. This concept supports speech recognition, pitch detection, and low-frequency sound processing.
Default mode		**Default Mode Network.** The default mode network includes medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, lateral parietal cortex, and medial temporal regions. It is commonly associated with internally oriented cognition, autobiographical memory, self-referential thought, and the integration of personal and social context.	**Precuneus.** The precuneus lies on the medial parietal cortex. It is commonly associated with autobiographical memory, self-related imagery, and mental context. This region is often described as supporting integration of personal and external information.	**Autobiographical Memory.** Autobiographical memory is commonly associated with the medial prefrontal cortex, posterior cingulate cortex, precuneus, angular gyrus, parahippocampal cortex, retrosplenial cortex, and lateral parietal cortex. This function supports remembering personal experiences, spatial context, and internally constructed events.
Visual		**Object Perception Network**	**Lateral Occipital Cortex**	**Visual Analysis**
Language		**Language Network**	**Middle Temporal Gyrus**	**Language Comprehension**
Somatomotor		**Somatomotor Network**	**Primary Somatosensory Cortex**	**Tactile Encoding**
Interoception		**Interoceptive Network**	**Anterior Insula**	**Interoceptive Processing**
Frontoparietal		**Frontoparietal Control Network**	**Inferior Frontal Gyrus**	**Working Memory**
Salience		**Salience Network**	**Anterior Cingulate Cortex**	**Executive Control**

Fig. R6. New Fig. 2. Auditory and default mode networks show the complete generated texts, and the other six their three generated titles. True is the atlas map. Maps show their top 5% of voxels in sagittal, coronal, and axial views. Supplementary Table S5 gives every text.

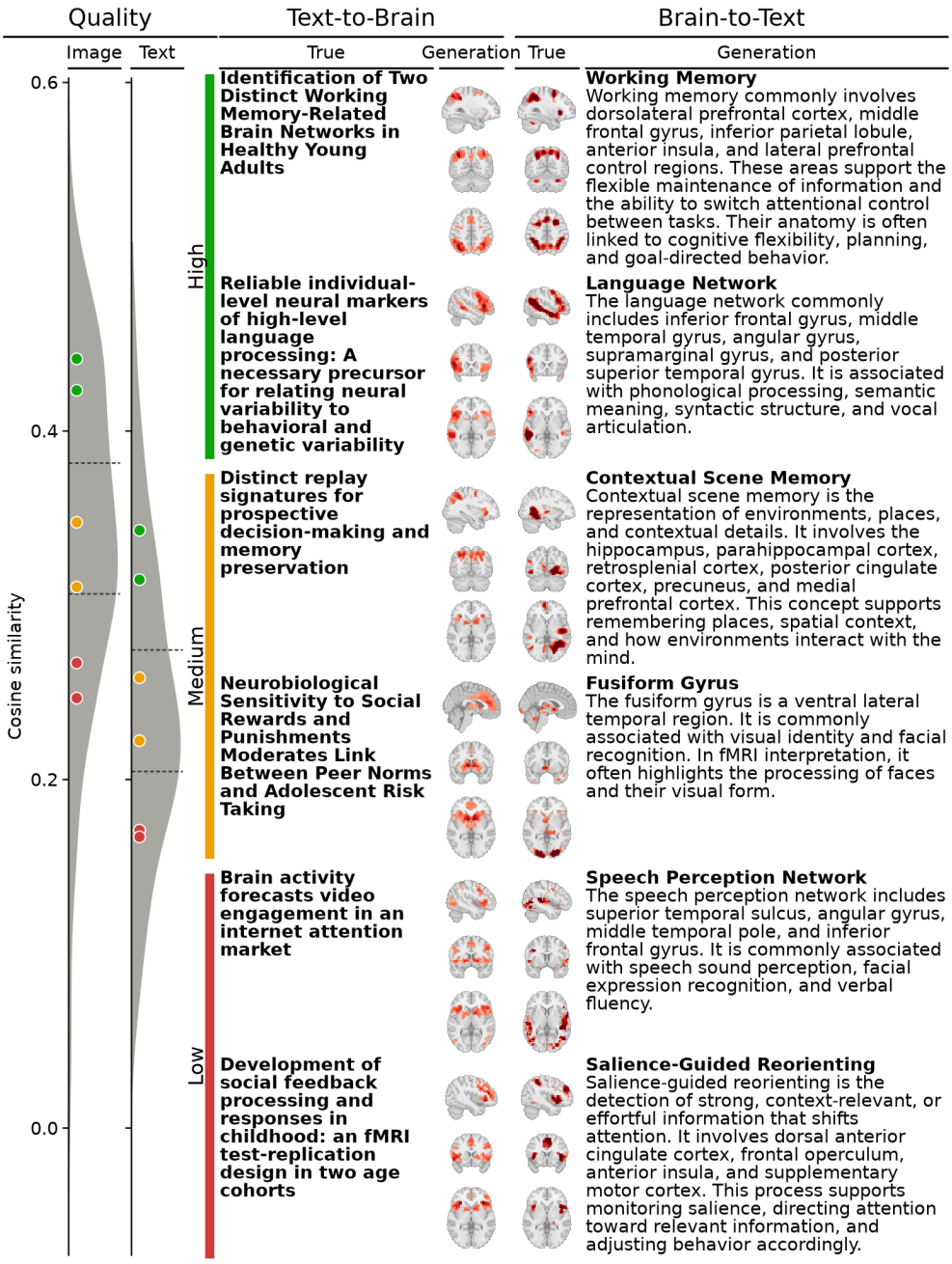


Fig. R7. New Fig. 3. NeuroVault papers from the high, medium, and low thirds of the shared-space cosines, chosen by a fixed rule (3.9). Low rows show a scattered true map (top) and a model miss on a clean map (bottom). Maps show their top 5% of voxels.



Fig. R8. New Fig. S4. Labeling of resting-state components from UK Biobank (top) and the Human Connectome Project (bottom). Bars give the five most similar labels from the contrastive models. The generated column is the [NETWORK] caption of the stage 2 MLP projector, its title in bold.

1.10: Overall assessment

Reviewer's comment

NeuroVLM is a creative and potentially useful framework, and the current results provide encouraging evidence that large collections of publication text and coordinate-derived maps can support cross-modal retrieval, broad functional labeling, and coarse generation. The manuscript would be considerably strengthened by independent validation of the generative outputs, clearer separation of datasets across all training stages, a revised NeuroVault evaluation, and more precise framing of the generated spatial representations. These revisions would help establish the appropriate scientific use cases and limitations of the framework.

Response: Comments 1.1, 1.2, 1.3, and 1.4 answer the four revisions named here, in the same order.

1.11: Code availability

Reviewer's comment

The repository is a useful community resource: it has an Apache-2.0 license, pip installation instructions, pretrained models and data, tutorials, tests, and CI across Python 3.10–3.13 on macOS and Ubuntu. The latest CI run passed 405 tests (one skipped), although reported code coverage was 47%.

The paper's results appear only partially reproducible. The repository includes data, training, and evaluation notebooks, but two preprocessing notebooks contain unresolved merge conflicts and are invalid; some analyses rely on manual steps, machine-specific paths, or external checkpoints; dependencies and Hugging Face artifacts are not fully pinned; and there is no figure-by-figure reproduction guide. Thus, the package appears usable for inference, but exact end-to-end reproduction would require additional fixes and documentation.

I reviewed the repository and existing CI results but did not install or run the code locally.

Response: We resolved the merge conflicts in both notebooks and added a figure-by-figure guide that names the script behind every paper figure. One command builds the pinned environment and fetches the Hugging Face inputs at pinned, checksummed revisions. It then retrains the model, redraws the figures, and checks each metric against the paper. Two runs from scratch gave byte-identical weights on the same GPU model and library versions.

Tests and coverage. Coverage is measured on the 57 files needed to train the models and draw the figures. The CNN and other experiments outside the paper are left out. Training takes hours, so continuous integration trains every model for one batch on CPU, which checks the whole pipeline end to end.

Remaining limits. Two inputs still come from outside the pinned set. They are the Automated Anatomical Labeling (AAL) atlas, whose download server fails, and the NiCLIP and NeuroConText outputs compared in Fig. S3.

Changes to the manuscript: New Data availability and Code availability sections after the Discussion, l. 954–986, replacing the links in Methods, Training, l. 185–190.

Reviewer 2

Reviewer's comment

The manuscript presents NeuroVLM, a unified framework for neuroimage–text contrastive learning, text-to-brain generation, and brain-to-text generation. The topic is meaningful, and the large-scale dataset provides potential value. However, the manuscript is not suitable for publication in its current form and several issues should be addressed.

Response: We addressed each issue (2.1 to 2.12).

2.1: Motivation of the Qformer

Reviewer's comment

1.The authors use a Q-Former for brain-to-text generation, but its motivation is unclear. Comparisons with simpler alternatives, such as an MLP projector or cross-attention module, should be provided.

Response: We agree. Our original Qformer attended over only two input tokens, the map latent and its semantic vector, so its attention had little to select. A multilayer perceptron (MLP) projector of the same size therefore replaces the Qformer in both stages. It raises stage 1 retrieval AUC from 0.82 to 0.90 (Fig. R9). A cross-attention module would also attend over only two tokens, so we did not add one.

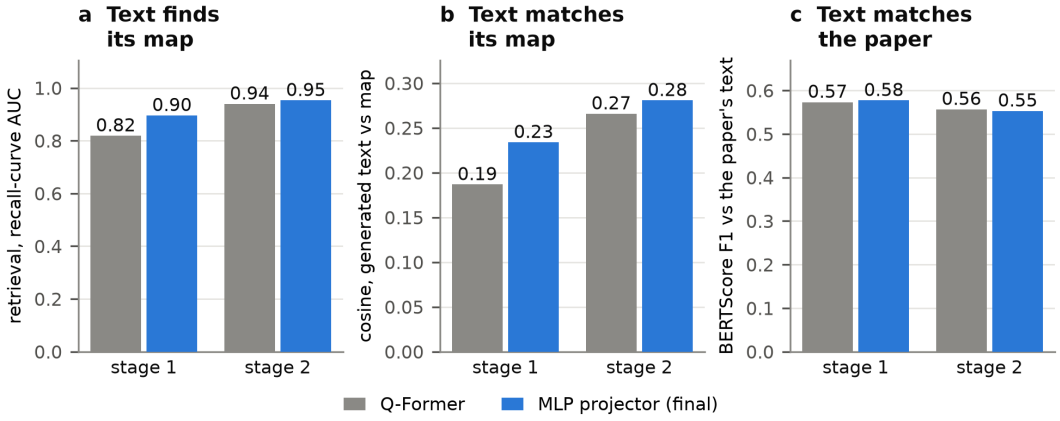


Fig. R9. Qformer and MLP projector. Gray bars show the original model with the Qformer and blue bars the final model with the MLP projector, on PubMed test maps. Panels show (a) retrieval AUC, (b) caption-to-map cosine, and (c) BERTScore F1 against the paper's text.

Changes to the manuscript: Methods, Text Decoder, l. 284–298, and Brain-to-Text, l. 341–362 (MLP projector). Results, Generative Models: Networks, l. 442–448 (projector named), and PubMed, l. 596–607. Supplementary Information, Brain-to-Text Projector, l. 1352–1374 and l. 1412–1428 (rewritten for the MLP projector), with Table S3 (p. 32). Figs. S1 and S4 captions (p. 23 and p. 34). New Supplementary Section Projector Comparison, l. 2073–2079, with Fig. S9 (p. 46).

2.2: Datasets and sample sizes per model

Reviewer's comment

2. The training data for the contrastive and generative models are not clearly described. The authors should specify the datasets and sample sizes used for each model and clarify whether the same image-text pairs were reused across different training stages.

Response: 1. Datasets and sample sizes. Fig. R3 and Table R7 give the data and sample size of each training stage (1.2).

2. Reuse across stages. Yes, the training papers are reused across stages. The autoencoder, the projection heads, and the stage 1 projector all learn from these papers. The anatomically centered head and the stage 2 projector learn from short concept texts, paired with maps built from training papers only (2.6). No validation or test paper trains these components. The language model's training script also removes them from its corpus (2.11).

Table R7. Training data per component. Training and validation sizes of each trained component of the final model, as in Fig. R3.

Training stage	Data	Train	Validation	Unit
Autoencoder	PubMed coordinate maps	23,894	2,987	papers
Text-to-brain and contrastive heads	SPECTER2 text and map latents	23,894	2,987	papers
Anatomically centered projection head	synthetic concept text–map pairs from training papers	6,309	64	pairs
Stage 1 projector (MLP)	map latents and LLM summaries	22,547	2,837	papers
Stage 2 projector (MLP)	[NETWORK], [REGION], [FUNCTION] texts (1,106 of 1,338 kept)	972	134	texts
NeuroQwen3-0.6B	PubMed titles and abstracts	1.22 M	12,317	articles

Changes to the manuscript: Methods, Training, l. 159–173. New Supplementary Section Data Flow and Training Details, l. 1745–1753, with Fig. S5 (p. 39) and Table S6 (p. 40), as for comment 1.2.

2.3: NeuroVault selection bias

Reviewer's comment

3. The NeuroVault evaluation has selection bias. When multiple images correspond to the same title and abstract, the contrastive model selects the most similar image. This procedure may substantially inflate the reported performance.

Response: We agree that choosing the most similar map inflates the score. That map is reported as an upper bound (1.3). A lower bound keeps each paper's worst-matching map, and a multi-target score counts any of the paper's maps (Fig. R4). Each rule has its own chance curve, from texts shuffled across papers. Under every rule, the model's AUC stays about 0.1 above chance.

Changes to the manuscript: Fig. 5b caption (p. 16). Results, Contrastive Model: NeuroVault, l. 754–767. Supplementary Information, NeuroVault preprocessing, l. 1177–1192 (map types and exclusions) and l. 1204–1210 (bounds). New Supplementary Section NeuroVault Bounds and Map Types, l. 1954–1963, with Fig. S7 and Table S9 (p. 44), as for comment 1.3.

2.4: Data leakage

Reviewer's comment

4.Potential data leakage across PubMed, Neurosynth Compose, and NeuroVault has not been sufficiently excluded. DOI-based filtering alone may not identify duplicate records, preprint and journal versions, or repeated coordinate tables.

Response: We audited leakage beyond DOI filtering. No DOI or PubMed identifier is shared across splits. The three checks below flag 16 test and 26 validation papers (Table R8). Every evaluation leaves them out. We also removed seven NeuroVault papers that duplicate a training paper.

1. Duplicate records. Each held-out paper was compared with the training and NeuroVault papers by fuzzy title matching and by the similarity of its title and abstract. Among the papers they flag is one paper stored as two records.

2. Preprint and journal versions. The same checks flag a preprint and its journal version, which carry different DOIs.

3. Repeated coordinate tables. The coordinate tables from PubMed and Neurosynth Compose were compared across all splits. Identical or near-identical tables mark a paper that repeats another paper's peaks.

Other training data. The synthetic maps of the anatomically centered head had pooled held-out papers, so we rebuilt them from training papers only. This fix and the removals each change AUC, Dice, and correlation by less than 0.01. The language model's training script also removes all evaluation papers from its corpus (2.11).

Table R8. Leakage audit. Held-out PubMed papers flagged by each check against other papers. Rows overlap, and the last row is their union.

Check against other papers	Test set (2,987)	Validation set (2,987)
Near-identical title	4	6
Near-identical title and abstract	3	6
Identical coordinate table	6	12
Near-identical coordinate table	12	21
Removed from every evaluation	16	26

Changes to the manuscript: Methods, Training, l. 169–181 (synthetic maps from training papers only, leakage audit, evaluation exclusions, and the evaluation papers removed from the language-model corpus). Methods, Datasets, l. 142–144 (NeuroVault papers that duplicate a training paper removed). Supplementary Information, NeuroVault preprocessing, l. 1177–1192.

2.5: Competitive baselines

Reviewer's comment

5.The manuscript lacks competitive baselines for the generative tasks. The proposed methods should be compared with existing approaches and simpler projection or retrieval-based baselines.

Response: Text to brain. Fig. R1 and Table R1 compare NeuroVLM's maps with existing approaches and simple baselines (1.1). NeuroVLM leads Neurosynth, NeuroQuery, mean and domain maps, and atlas labeling. It narrowly leads linear regression, the simple projection baseline, and

leads the nearest training paper's map, the retrieval baseline. Text2Brain, a published model, trails NeuroVLM (Dice 0.49 against 0.53), most on papers outside its training corpus.

Brain to text. We know of no published model that writes text from group-level brain maps. Existing decoders NiCLIP and NeuroConText rank fixed texts, and NeuroVLM is slightly ahead of both (Fig. S3). A simpler MLP projector beats the Qformer and replaces it (2.1). As a retrieval baseline, the nearest training caption is no closer to the paper's text than generated captions, and it carries details of other studies (3.3).

Changes to the manuscript: Methods, Text-to-Brain, l. 365–385 (final head), and Performance Metrics, l. 398–404 and l. 415–426. Results, Generative Models: PubMed, l. 596–607, and Fig. 4 caption (p. 15). New Supplementary Section Text-to-Brain Baselines and Retrieval, l. 1843–1859, with Fig. S6 (p. 42) and Table S8 (p. 43), as for comment 1.1.

2.6: Summarized and synthetic text

Reviewer's comment

6.The construction of the summarized and synthetic text data is unclear. The authors should describe the generation procedure and provide representative examples.

Response: One prompt generated the paper summaries and another generated the synthetic and curated texts. Both prompts are given below, and Table R9 gives a complete example of each kind of text.

Summaries. Llama-3.3-70B-Instruct summarized each paper with the first prompt as its system message, removing low-level study details. The user message was the paper's title and abstract, joined by a separator token. Decoding sampled with temperature 0.3 and top-p 0.9, so a rerun words each summary differently. Summaries re-created for test-set papers cover the same topics as the released ones. The stage 1 projector learns from these summaries.

Synthetic texts. Qwen3.5-27B wrote short entries with the second prompt. The seeds were region names from standard brain atlases, Cognitive Atlas concepts, and the network names of our network atlases. The user message was the seed alone, such as "Insula". Decoding used temperature 0.1 and top-p 0.9, with thinking mode off. Sampling each seed several times gave re-wordings of one concept. The anatomically centered head learns from these entries.

Curated texts. The stage 2 projector learns from 1,338 curated texts about networks, regions, and functions. Qwen3.5-27B wrote them in the same way. The original generation code was not kept. A script in the code repository repeats the process, and its texts match the released texts of the same seeds in content (MiniLM cosine 0.88 on average).

```
Model: meta-llama/Llama-3.3-70B-Instruct
Decoding: sampling, temperature 0.3, top-p 0.9, top-k 50, at most 256 new tokens
User message: title[SEP]abstract
System message:

You are writing a short, topic-focused neuroscience mini-review.

INPUT:
You will receive:
    title + " [SEP] " + abstract from an fMRI publication.

GOAL:
Write a single coherent paragraph (4-6 sentences) that:
- Describes the scientific topic implied by the paper (e.g., meditation and the default mode
  network, exercise and working memory), not the paper's methods.
- Explains the key cognitive processes (e.g., working memory, self-referential thought,
  emotion regulation) and the main brain systems or networks involved.
- May mention typical effects or relationships in general terms (e.g., ‘‘has been associated
  with changes in connectivity'' or ‘‘can influence executive control''), but should stay high-level.

ALLOWED:
- General statements about effects or relationships (e.g., ‘‘exercise has been associated
  with improved cognition'', ‘‘meditation can alter DMN connectivity'').
- Generic references to ‘‘studies'' or ‘‘research'' in the field.
- Qualitative descriptions of brain activity or connectivity (e.g., ‘‘increased activation
  in prefrontal regions'', ‘‘changes in functional connectivity'').

AVOID:
- Exact numerical details (sample sizes, p-values, coordinates, time points).
- Detailed task or protocol descriptions from the specific study.
- Directly quoting or closely paraphrasing long sentences from the abstract.
- Overly narrow focus on this single experiment; write as if summarizing the topic area.

STYLE:
- Neutral, scientific tone, like a short review or encyclopedia entry.
- One paragraph only. No bullet points or headings.
- Focus on giving a clear sense of ‘‘what this topic is about'' in the context of the field.
```

```
Write a short educational neuroscience entry for the provided fMRI concept.

Requirements:
- Explain the concept in general neuroscience terms.
- Always include relevant anatomical regions.
- Connect the anatomy to the cognitive or neural function.
- Use cautious, broad language; avoid overclaiming.
- Do not mention studies, patients, diseases, disorders, treatments, age groups, tasks, results,
  methods, or statistics.
- Do not use: critical for, responsible for, causal, proven, this study, activation, altered,
  reduced, increased.
- Do not invent highly specific subregions unless they are commonly associated with the concept.
- If the concept is broad, name the main commonly associated regions or networks.
- If the concept is anatomical, describe its broader network/function associations.
- Write 3-5 complete sentences.

Return valid JSON only:
{
  "title": "1-5 words",
  "description": "3-5 sentences"
}
```

Table R9. Examples of the training texts. Each row gives one complete text and the training stage that uses it. Llama-3.3-70B-Instruct wrote the summaries and Qwen3.5-27B the synthetic and curated texts.

Text (used by)	Complete example
Summary of a training paper (stage 1 projector)	The human parietal and opercular regions are involved in processing somatosensory information. Somatotopy refers to the organization of these regions according to the location of sensory inputs on the body. The parietal cortex, including areas such as the postcentral gyrus and intraparietal sulcus, contains multiple somatosensory areas with distinct somatotopic representations. Attentional modulation can enhance activity in these areas, particularly in higher-order regions.
Synthetic text "Insula" (anatomically centered head)	The insula is a deep cortical region situated within the lateral sulcus of the cerebral hemisphere. This area maintains connections with structures such as the anterior cingulate cortex and the amygdala. Processes occurring in this region are often linked to perceptions of internal bodily states and emotional awareness. It contributes to the integration of sensory information with subjective experience across various brain networks.
Curated [NETWORK] text "Salience Network" (stage 2 projector)	The salience network commonly includes anterior insula and dorsal anterior cingulate cortex. It is associated with detecting and prioritizing behaviorally relevant internal and external information. Its connections with limbic, autonomic, and frontoparietal systems support flexible shifts between internal monitoring and externally oriented control.

Changes to the manuscript: Methods, Brain-to-Text, l. 341–362. Supplementary Information, Text preprocessing, l. 1095–1174 (both prompts, their models, seeds, and settings, and the examples).

2.7: Dataset sizes

Reviewer's comment

1.The reported dataset sizes are inconsistent. The Methods section reports 3,000 NeuroVault images and 100 network images, whereas other sections report 4,553 and 300, respectively.

Response: As for comment 1.5, we recomputed every count from the released data and corrected the text and captions (Table R5). The supplement's 4,553 maps predate image quality control and the removal of papers already in PubMed. Neither network-image count can be reproduced, so the released counts replace both.

Changes to the manuscript: Methods, Datasets, l. 132–144. Results, Contrastive Model: Network Labeling, l. 735–747, and Autoencoder, l. 826–830. Figs. 4 and 5 captions (pp. 15–16). Supplementary Information, NeuroVault preprocessing, l. 1177–1192 and l. 1209–1211, as for comment 1.5.

2.8: Fig. 5 caption

Reviewer's comment

2.The caption of Fig. 5 reports both 0.78 and 0.85 for the NeuroVault AUC.

Response: The corrected caption names the curve behind each value. With each paper's best-matching map, the AUC is 0.85 from text to brain and 0.79 from brain to text (Fig. R10b). P-maps are inverted, so that high values mark strong effects, as for comment 1.3. Each curve has its own dashed chance curve, from texts shuffled across papers.

Lower bound. The lower bound and its chance curve lie below 0.50 by construction. The lower bound keeps the map that the model scores lowest for each paper. That map ranks below a typical candidate even when the texts are shuffled. The model still beats this chance level.

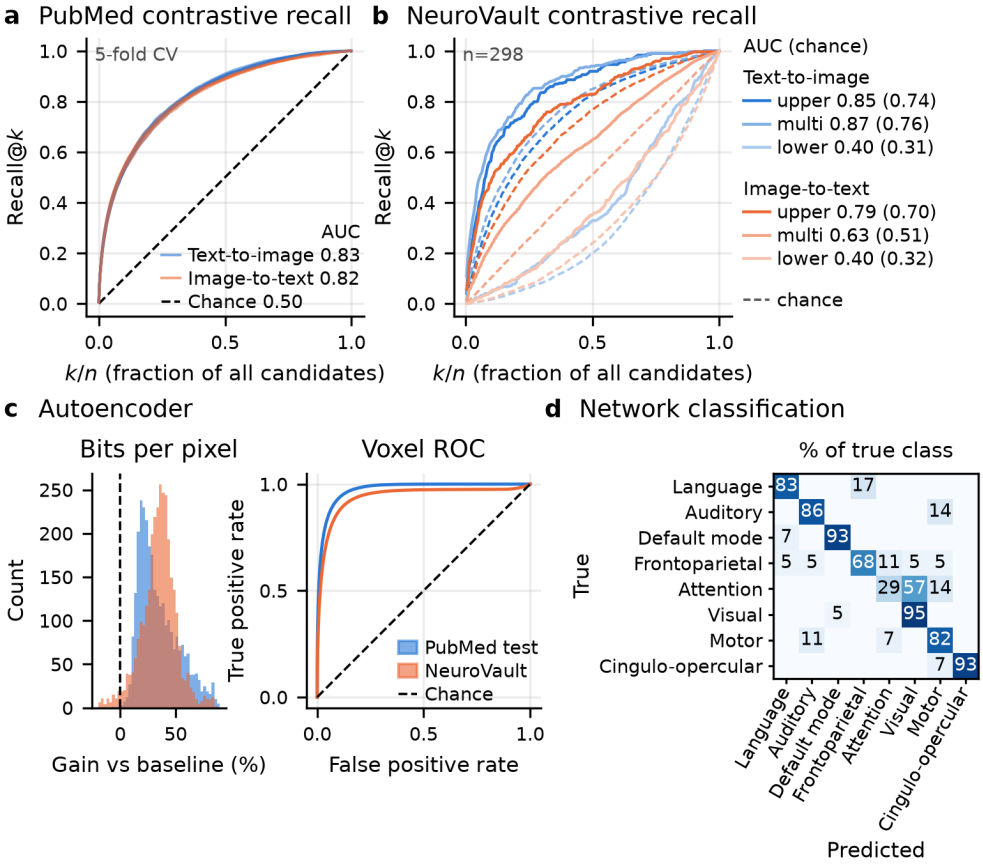


Fig. R10. New Fig. 5. Contrastive recall on (a) PubMed and (b) NeuroVault, (c) autoencoder reconstruction, and (d) network classification. In (b), each paper's target is its best map (upper bound), every map (multi-target), or its worst map (lower bound), and dashed lines show chance.

Changes to the manuscript: Fig. 5 caption (p. 16). Results, Contrastive Model: NeuroVault, l. 754–767, and PubMed, l. 771–780.

2.9: SPECTER naming

Reviewer's comment

3. The use of "SPECTER," "SPECTER2," and "Specter" is inconsistent. The exact model used in each component should be clarified.

Response: Every text-encoding step uses SPECTER2, the allenai/specter2_aug2023refresh model with its ad hoc query adapter (1.6). The original SPECTER is never used. We corrected every mention to SPECTER2. The language model is NeuroQwen3-0.6B, which is Qwen3-0.6B adapted to PubMed abstracts (2.11). Llama-3.3-70B-Instruct wrote the paper summaries, and Qwen3.5-27B wrote the synthetic and curated texts (2.6).

Changes to the manuscript: Introduction, l. 107–111. Methods, Text Encoder, l. 220–227, and Text-to-Brain, l. 365–373. Supplementary Information, Text preprocessing, l. 1085–1088. Fig. S1 caption (p. 23), as for comment 1.6.

2.10: LoRA analogy

Reviewer's comment

4.In Section 2.9.2, describing an MLP projection head as similar to a LoRA adapter is technically inaccurate.

Response: We agree and deleted the LoRA analogy (1.7). The head is a small network trained on frozen SPECTER2 embeddings. The revision calls it an anatomically centered projection head, not an adapter.

Changes to the manuscript: Methods, Text-to-Brain, l. 365–385 (LoRA sentence deleted, projection head named). Supplementary Information, Projection Head, l. 1272–1294, as for comment 1.7.

2.11: Training details

Reviewer's comment

5.The authors should report the model parameters, GPU hardware, training time, random seeds, Qwen fine-tuning strategy, and relevant training hyperparameters.

Response: Table R10, a new Supplementary Table, gives the parameters, training data, epochs, optimizer, learning rate, batch size, seed, and training time of every component. All components except the language model train on one RTX 5090 GPU, with fixed seeds and deterministic kernels. The whole pipeline, five-fold cross-validation included, trains from scratch in about 1.8 hours.

Qwen fine-tuning strategy. NeuroQwen3-0.6B is Qwen3-0.6B with all weights fine-tuned, without adapters, on PubMed titles and abstracts (Table R10). It trained for one epoch on one H100 GPU, and its training time was not recorded. Its training script removes the validation, test, and NeuroVault evaluation papers from this corpus. Comment 1.2 reports a memorization check.

Table R10. Training details. Rows give the trainable parameters, training data, epochs, optimizer settings, seed, and training time of each component of the final model. Times are for one RTX 5090 GPU, except for NeuroQwen3-0.6B.

Component	Parameters	Training data (train / validation)	Epochs (selection)	Optimizer, learning rate, batch	Seed	Time
Autoencoder	59.9 M	PubMed maps, 23,894 / 2,987	101 (lowest validation loss)	AdamW, 5e-5, 256	0	10 min
Contrastive heads, text and image	0.59 M and 0.30 M	text and map latents, 23,894 / 2,987	301 (last)	AdamW, 1e-5, 2,048	123	6 s
Text-to-brain head	0.59 M	text and map latents, 23,894 / 2,961	30 of at most 80 (best validation Dice)	AdamW, 1e-3, 512	0	22 s
Anatomically centered head	3.8 M	synthetic pairs, 6,309 / 64, and PubMed training pairs	200 of 2,000 (best PubMed validation recall)	AdamW, 2e-4, 2,048	123	103 s
Stage 1 projector (MLP)	28.3 M	maps and LLM summaries, 22,547 / 2,837	4 of at most 30 (lowest validation loss, early stopping)	AdamW, 3e-4 with cosine decay, 32	17	17 min
Stage 2 projector (MLP)	28.3 M	concept texts and maps, 972 / 134	30 (last)	AdamW, 1e-4 with cosine decay, 18	3 (best of 1–5)	5 min per seed
NeuroQwen3-0.6B	596 M	PubMed titles and abstracts, 1.22 M / 12,317 (385 M tokens)	1	AdamW, 5e-6 with cosine decay, 16 sequences of 1,024 tokens, bf16	12345	not recorded (one H100)

Changes to the manuscript: Methods, Training, l. 174–190 (exclusions, the evaluation papers removed from the language-model corpus, and training time). Methods, Text Decoder, l. 274–283 (NeuroQwen3-0.6B training). Supplementary Information, Autoencoder training, l. 1265–1268. New Supplementary Table S7 with the training details per component (p. 40).

2.12: Random results and failure cases

Reviewer's comment

6.Figs. 2 and 3 mainly show selected qualitative examples. Randomly selected results and representative failure cases should also be provided.

Response: We added random results and failure cases for every evaluation set. They include random test papers, the worst test-set cases in both directions, the weakest atlas networks, and NeuroVault papers that fail on their best-matching map. Failures are not rare, since 16% of test-set captions have a negative similarity to the paper's text. Fig. R11 shows the six worst brain-to-text cases. The other five sets are in the code repository.

Invented names. With unsteered title decoding, 23% of test-set [NETWORK] titles and 9% of [REGION] titles matched no canonical or atlas name, such as "Cerebro-Cervico-Brain". The language model still generates every title, but it is steered toward the canonical names. At the title position, tokens that cannot continue a canonical network or region name are masked, and the model's own probabilities choose the name. Nothing is classified or looked up.

Body text. The body text and [FUNCTION] titles are generated without this mask. They can still contain invented words or stray fragments, such as "useary" in Fig. R11, a limitation the Discussion

states.

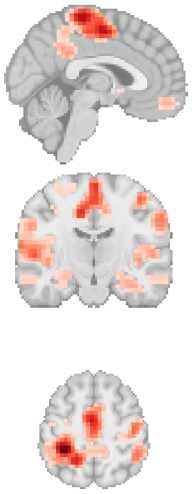
Non-Latin text. A few captions contained non-Latin characters from the language model's multilingual vocabulary. Decoding uses Latin-script tokens only, which removes them.

Changes to the manuscript: Figs. 2, 3, and S4 (p. 11, p. 12, and p. 34), with the Low rows of Fig. 3 as failure cases. Methods, Text Decoder, l. 299–306 (title decoding steered toward the canonical names, and Latin-script decoding). Supplementary Information, Brain-to-Text Projector, l. 1406–1413. Discussion, limitations, l. 899–905 (failure rate, typical failures, and invented words or stray fragments in the body text).

PubMed test set: the 6 worst brain-to-text cases (1-2 of 6)
The 6 test maps whose generated description is least similar to the paper's own text (title + summary; cosine in the shared text-brain space), out of all PubMed test maps. Maps: top 10% of voxels; texts complete.

1. PMID 27592810

true map



brain to text: cosine -0.713 (percentile 0.02 of 2,971)

Paper text (title + summary)

Retrieved emotional context influences hippocampal involvement during recognition of neutral memories.
During memory recognition, the emotional context in which a neutral memory is encoded can influence its retrieval and involvement of the hippocampus. The hippocampus, which plays a crucial role in memory formation and consolidation, interacts with the amygdala, a structure involved in emotional processing, to facilitate the encoding and retrieval of emotionally arousing experiences. When an emotionally arousing context is retrieved, the hippocampus, particularly an area adjacent to the amygdala, exhibits heightened activation and increased functional connectivity with the parahippocampal gyrus, a region involved in processing visual information. This interaction between the hippocampus and amygdala can lead to confident judgments of recognition for neutral memories, even after a delay. The retrieval of an emotionally arousing context can thus shape the memory of neutral events, influencing their recognition and recall. This highlights the dynamic nature of memory and its susceptibility to the emotional context in which it is retrieved.

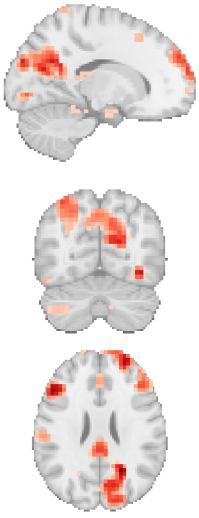
Generated description (complete)

[NETWORK] Sensorimotor Network The sensorimotor network includes precentral and postcentral gyri, as well as supplementary motor area. It is broadly tied to the execution of movement-related actions with high temporal precision. The network is often described as a tightly coupled system for bodily sensation and control. Its hubs are thought to support coordinated feedback from body parts during action planning or execution. **[REGION] Postcentral Gyrus** The postcentral gyrus is a primary somatosensory region in the brain. It contains cortical regions that support tactile and proprioceptive information processing related to body contact.
describe when it was used.
useary: When describing bodily sensorimotor representations, this region often serves as an intermediate hub for touch-related signals from fingertips or other skin surfaces of the hand or foot. **[FUNCTION] Body Sensory Mapping** Body sensory mapping is the cortical representation of body parts and their interactions with touch, pressure, proprioception, vibration, and temperature. It involves primary somatosensory cortex in regions such as parietal lobule 4B (PLC-4b), posterior insula, premotor cortex, thalamus, and supplementary motor area. This concept supports tactile navigation, spatial awareness, coordination between senses, and integration into self-motion perception.

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2. PMID 21715612

true map



brain to text: cosine -0.671 (percentile 0.05 of 2,971)

Paper text (title + summary)

Reciprocal anatomical relationship between primary sensory and prefrontal cortices in the human brain.
In the human brain, there is a reciprocal anatomical relationship between primary sensory cortices, such as those responsible for processing visual and auditory information, and the prefrontal cortex, a region involved in executive functions and decision-making. The gray matter volume of primary sensory cortices, including the visual and auditory cortices, is inversely correlated with the gray matter volume of the anterior prefrontal cortex. This means that as one area increases in size, the other decreases, forming a trade-off between the two regions. This relationship is observed along distinct anatomical pathways, with the dorsal and ventral parts of primary sensory cortices showing anticorrelation with the dorsolateral and ventromedial parts of the anterior prefrontal cortex, respectively. This anatomical trade-off suggests that primary sensory cortices and the anterior prefrontal cortex may be influenced by shared genetic or developmental factors, challenging the traditional view that brain area sizes simply scale with overall brain size.

Generated description (complete)

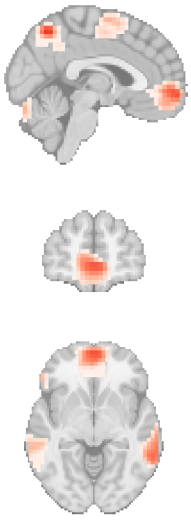
[NETWORK] Semantic Network The semantic network includes medial temporal cortex, hippocampus, anterior cingulate gyrus, and posterior parietal lobule. It is associated with internally-focused representations of the world—how things are connected or arranged in space. Its connections involve prefrontal cortex, amygdala, basal ganglia, retrosplenial cortex, and lateral occipital regions. This network helps organize internalized meaning into external experiences for processing.
[REGION] Medial Prefrontal Cortex Broadly tied to the processing of emotion and value, this region is often engaged during deliberation over personal preferences. It is associated with internal thought about mood or behavior. Its activity tends to accompany reflective evaluation rather than direct action. It is generally linked to self-awareness and long-term memory.
[FUNCTION] Memory Processing Memory processing commonly involves the hippocampus, amygdala, retrosplenial cortex, parahippocampal gyration, medial prefrontal cortex, and lateral temporal lobe. These regions support encoding of event-related information, retrieval of contextual memory components, integration with affective states, spatial context for episodic events, and semantic knowledge about objects or concepts.

Fig. R11. Worst brain-to-text cases. The six PubMed test maps whose caption is least similar to the paper's text, two per page. Each shows the map in sagittal, coronal, and axial views, the paper's title and summary beside the complete generated caption, and the caption's similarity with its percentile. (Page 1 of 3.)

PubMed test set: the 6 worst brain-to-text cases (3-4 of 6)

3. PMID 26485148

true map



brain to text: cosine -0.585 (percentile 0.08 of 2,971)

Paper text (title + summary)

Human-Robot Interaction: Does Robotic Guidance Force Affect Gait-Related Brain Dynamics during Robot-Assisted Treadmill Walking? During robot-assisted treadmill walking, the brain's sensorimotor cortex plays a crucial role in locomotor control. This region exhibits gait-related spectral modulations in the mu, beta, and low gamma bands, which are associated with specific phases of the gait cycle. When a robotic device provides assistance, such as the Lokomat gait orthosis, it can affect the brain's activity patterns, particularly in the right primary sensory cortex. Specifically, mu and beta rhythms are suppressed in this area during robot-assisted walking with high guidance force, indicating a reduced involvement of the sensorimotor area. In contrast, a lower level of guidance force allows for a more active participation of the sensorimotor cortex, which is essential for motor learning. This suggests that optimal training parameters for robot-assisted treadmill walking should minimize the use of high guidance force to maximize the engagement of the sensorimotor cortex.

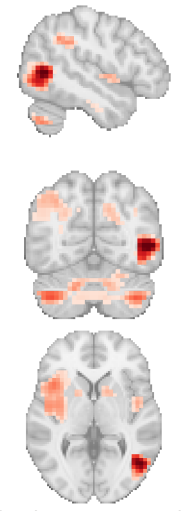
Generated description (complete)

[NETWORK] Language Network The language network includes inferior frontal cortex, middle temporal gyrus, posterior cingulate layer and lateral occipital cortex. It is broadly associated with comprehension of words, meaning, speech production, linguistic structure, semantics, and conceptual integration across domains. Its distributed organization supports the flexible mapping between semantic knowledge and phonological processing.
[REGION] Middle Frontal Gyrus The middle frontal gyrus is a large lateral region in the brain. It commonly involves tasks related to language and conceptual understanding. In fMRI, this region is often associated with controlled selection of information from memory or speech-related contexts.
[FUNCTION] Semantic Processing Sensory processing commonly includes anterior temporal cortex, middle frontal regions, and inferior parietal areas. These regions support meaning-based retrieval of words or concepts from stored knowledge. Semantic analysis often involves ventral visual word system, posterior medial prefrontal cortex for conceptual integration, and superior temporal gyrus for linguistic mapping.

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4. PMID 32595302

true map



brain to text: cosine -0.583 (percentile 0.12 of 2,971)

Paper text (title + summary)

Autism and Beauty: Neural Correlates of Aesthetic Experiences in Autism Spectrum Disorder In individuals with Autism Spectrum Disorder (ASD), the neural correlates of aesthetic experiences differ from those of typical individuals. When performing aesthetic judgments, individuals with ASD exhibit altered activity in key brain regions, including the temporoparietal area and insula, which are involved in processing and integrating sensory information. In contrast, the anterior region of the superior frontal gyrus, a region associated with executive functions and decision-making, shows reduced activity in individuals with ASD. Additionally, the amygdala, a structure involved in emotional processing, and the posterior region of the middle/inferior temporal gyrus, a region involved in visual processing, exhibit increased activity in individuals with ASD during aesthetic judgments of fractal images. These findings suggest that individuals with ASD may process and experience aesthetic stimuli in distinct ways compared to typical individuals.

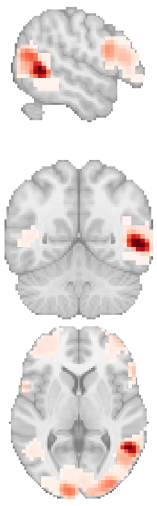
Generated description (complete)

[NETWORK] Cerebro-Cerebellar Network The cerebrally located network includes the lateral parietal cortex, prefrontal lobes, anterior cingulate gyrus, and posterior cingulum. It is commonly associated with coordination of action across time frames (e.g., timing), prediction based on current information, cognitive control over movement execution, regulation of attention span, and interoceptive awareness.
[REGION] Dentate Nucleus The dentate nucleus is a lateral region of the brain located near the cerebellum. It is associated with motor learning and maintenance, as well as cognitive functions related to action planning. Its anatomy links movement patterns with attentional control.
[FUNCTION] Predictive Coordination Predictive coordination is the integration of anticipatory and current action to optimize performance. It involves cerebellar vermis, dentate nucleus, thalamus, globus pallidum, ventral tegmental area, parietal cortex, prefrontal regions, motor cortex, sensory cortices, basal ganglia, and association cortex. This concept supports prediction-based decision-making across cognitive domains.

PubMed test set: the 6 worst brain-to-text cases (5-6 of 6)

5. PMID 31866810

true map



brain to text: cosine -0.577 (percentile 0.15 of 2,971)

Paper text (title + summary)

Chronic Tinnitus Exhibits Bidirectional Functional Dysconnectivity in Frontostriatal Circuit Chronic tinnitus is associated with abnormal functional coupling between the nucleus accumbens (NAc) and the prefrontal cortex, particularly the middle frontal gyrus (MFG), orbitofrontal cortex (OFC), and inferior frontal gyrus (IFG). This bidirectional dysconnectivity in the frontostriatal circuit may contribute to the evaluation and modulation of sensory signals. In individuals with chronic tinnitus, the NAc exhibits aberrant directional connectivity with the prefrontal cortex, which is linked to tinnitus characteristics, including distress and duration. Specifically, increased connectivity from the right NAc to the left MFG and from the right MFG to the left NAc is positively correlated with tinnitus handicap, while enhanced connectivity from the right NAc to the right OFC is associated with longer tinnitus duration. These findings suggest that the frontostriatal circuit plays a crucial role in the neural basis of tinnitus sensation and may provide insights into the neuropathological features of the condition.

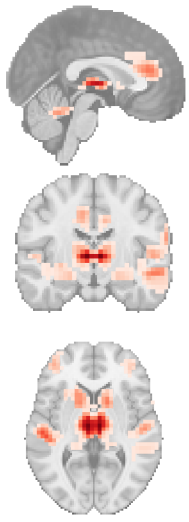
Generated description (complete)

[NETWORK] Ventral Attention Network The ventral attention network includes the frontal eye field, dorsal frontoparietal cortex, and lateral occipital regions. It is broadly associated with spatially guided attention. Its anatomy links to how visual information is processed in a coordinated manner for effective foveal vision during saccadic motion detection. The circuit supports both top-down directed attention and bottom-up involuntary responses to visually salient stimuli.
[REGION] Lateral Occipital Cortex The lateral occipital cortex is a region of ventral visual pathways. It supports analysis of shape and form, describing how objects appear in space. In general, it occupies the dorsal surface of inferior temporal sulcus.
[FUNCTION] Motion Perception Motion perception is the ability to detect and interpret movement in space. It involves motion-related visual cortex along with inferior frontal areas for downstream analysis of spatial relationships between objects or events. The network includes dominant occipital, intraparietal, superior temporal, dorsal frontoparietal, ventral premotor, and lateral prefrontal regions. Its anatomy supports both object-based and event-driven understanding of spatial information.

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6. PMID 29542095

true map



brain to text: cosine -0.576 (percentile 0.19 of 2,971)

Paper text (title + summary)

Tuning down the hedonic brain: Cognitive load reduces neural responses to high-calorie food pictures in the nucleus accumbens. When individuals are under cognitive load, neural responses to high-calorie food pictures in the brain's reward system, specifically the nucleus accumbens, are reduced. This region is involved in processing rewarding and pleasurable stimuli, such as food. Cognitive load, which can be induced by tasks requiring mental effort, appears to dampen the brain's response to high-calorie food images, potentially reducing their appeal. This effect may be related to the brain's ability to prioritize tasks and allocate resources, leading to a decrease in the neural processing of distracting stimuli like high-calorie food. The reduction in neural responses to high-calorie food pictures in the nucleus accumbens may contribute to a decrease in food cravings and consumption.

Generated description (complete)

[NETWORK] Auditory Network The auditory network includes primary and secondary complex acoustic networks, supported by the superior temporal gyrus. It is broadly tied to pitch, rhythm, timbre, speech perception, music processing, and sensorimotor sound experience. Its anatomy spans Heschl region, inferior frontal regions for feature association, parietal cortex for spatial organization of sounds, and posterior central sulcus areas involved in holistic perceptual systems.
[REGION] Brainstem This region is broadly associated with the regulation of arousal and the modulation of basic physiological processes. It is often linked to wakefulness, attentional control, and fundamental brain functions such as sleep-wake cycles. Its activity tends to correlate with general state transitions in the nervous system.
[FUNCTION] Early Auditory Processing This function supports the perception of basic auditory signals, such as sound waves and natural environmental sounds. It is associated with primary auditory cortex in temporal-limbic regions around the superior temporal gyrus and frontal-associated belt areas. This representation broadens the cortical basis for early acoustic signal processing.

Reviewer 3

Reviewer's comment

Hammonds et al. introduce a meta-analysis framework for aligning text-neuroimage pairs in a shared latent space. The framework can retrieve one modality from the other by comparing and ranking the latent vector embeddings. The framework can also generate one modality from the other by passing the latent vector embeddings through generative text and brain decoders. The authors tested the framework on multiple neuroimaging datasets. They found statistically significant retrieval and generation performance with sensible qualitative examples.

Response: No change was requested.

3.1: Novelty and what the model learns

Reviewer's comment

Augmenting existing meta-analysis tools with the flexibility provided by LLMs is an important and promising research direction. The manuscript appears technically sound and the NeuroVLM architecture seems appropriate for integrating the contrastive and generative models. However it is less clear how NeuroVLM compares to previous meta-analysis methods. The manuscript would benefit from a) clarifying the novelty of the framework by demonstrating the importance of generation and context and b) interpreting what the model learns.

Response: 1. Generation and context. Generated maps match each test paper's map better than the map of the nearest training paper (1.1). By MiniLM cosine, generated captions are as close to the paper's text as retrieved captions. Retrieved captions often report details of another study, and generated captions rarely do (3.3). Context also matters, since SPECTER2 beats a context-free word2vec encoder on every measure, by a modest margin (3.4).

2. What the model learns. Decoded latent dimensions match the Neurosynth maps of their top terms (3.5). Swapping one word moves the generated map toward the matching Neurosynth contrast, but negation has no effect. The cross-entropy loss matters most for retrieval.

Changes to the manuscript: Introduction, l. 99–105, Results, Generative Models: PubMed, l. 608–615, and Discussion, l. 863–868, which state the contribution of generation and context.

3.2: Comparison with existing methods

Reviewer's comment

My main concern is about the scope of the manuscript. NeuroVLM is clearly a useful tool but it is difficult to evaluate how much it improves on previous work. If the main contribution is higher performance then it would be helpful to include more direct comparisons. The only comparison between NeuroVLM and existing methods is in Figure S3 where it had comparable performance to NiCLIP and NeuroConText (albeit with fewer parameters). If the main contribution is instead more capabilities then it would be helpful to emphasize what NeuroVLM can do that previous methods cannot.

Response: 1. Direct comparisons. NeuroVLM gives the best text-to-brain maps, narrowly ahead of linear regression and ahead of Text2Brain, NeuroQuery, and Neurosynth (Fig. R1). Each method turns a test paper's title and abstract into a map, which is scored against that paper's map (Dice 0.53 against at most 0.49). In retrieval, NeuroVLM is slightly ahead of NiCLIP and NeuroConText on test papers that neither was trained on (Fig. R12).

2. New capabilities. NeuroVLM is, to our knowledge, the only model that writes text from group-level brain maps. It retrieves and generates in both directions through one shared space. Its generated maps and captions match or beat retrieved ones (3.3). The Introduction emphasizes these capabilities.

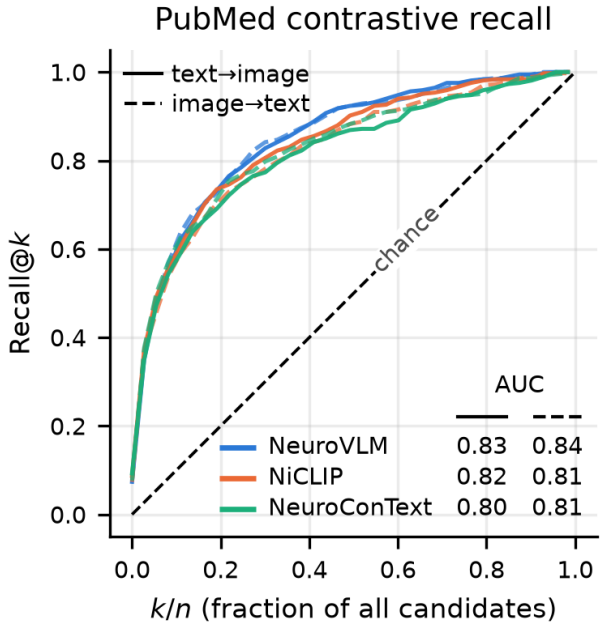


Fig. R12. New Fig. S3. Recall@k for NeuroVLM (blue), NiCLIP (orange), and NeuroConText (green) on the 366 PubMed test papers that none of the three models was trained on. Solid lines are text to image and dashed lines image to text.

Changes to the manuscript: Introduction, l. 99–105, on the new capabilities. Results, Generative Models: PubMed, l. 608–615, and Contrastive Model: PubMed, l. 771–780. Supplementary Information, Model Comparison, with Fig. S3 redrawn from the final model (p. 33). New Supplementary Section Text-to-Brain Baselines and Retrieval, l. 1843–1859, with Fig. S6 (p. 42).

3.3: Generation versus retrieval and granularity

Reviewer's comment

One capability that sets NeuroVLM apart from previous methods is that it can generate original text and neuroimages rather than only retrieving from a fixed set. However it would be helpful to show that the generated outputs are in fact more accurate and useful than retrieved outputs. For example if NeuroVLM uses an overfit decoder that collapses into a fixed set of outputs then it might be no better than existing retrieval methods. In fact the canonical projection matrix might explicitly drive the framework towards more retrieval-like outputs. Is there a way to quantify the granularity of NeuroVLM based on its output space?

Response: 1. Generated versus retrieved outputs. Generated maps are closer to each test paper's map than retrieved maps (1.1). By MiniLM cosine, generated captions are at least as close to the paper's text as the nearest training caption (Table R11). BERTScore favors training captions, even random ones, because they share the reference's summary style. Retrieved captions often carry another study's details, and generated captions rarely do (57% against 2%).

2. Collapse. The decoder does not collapse to a fixed set of outputs, since over 99% of test captions are unique (Fig. R13). The language model generates every caption. Network and region

titles are steered to canonical names by design, by masking tokens that cannot continue one (2.12). The body text is not masked.

3. Canonical projection. The canonical projection does pull captions toward its prototypes, as the reviewer suspected. With it, about one caption in ten nearly copies a curated training text, against almost none without it. We added this trade-off to the Methods.

4. Granularity. In the output space, test captions use nearly all canonical network and region names, and no title labels more than one test map in ten. The share of captions naming a study detail, such as a sample size, patient group, or task, falls from 36% at stage 1 to 2% at stage 2 (Fig. R13). Stage 2 captions describe the map rather than a single study.

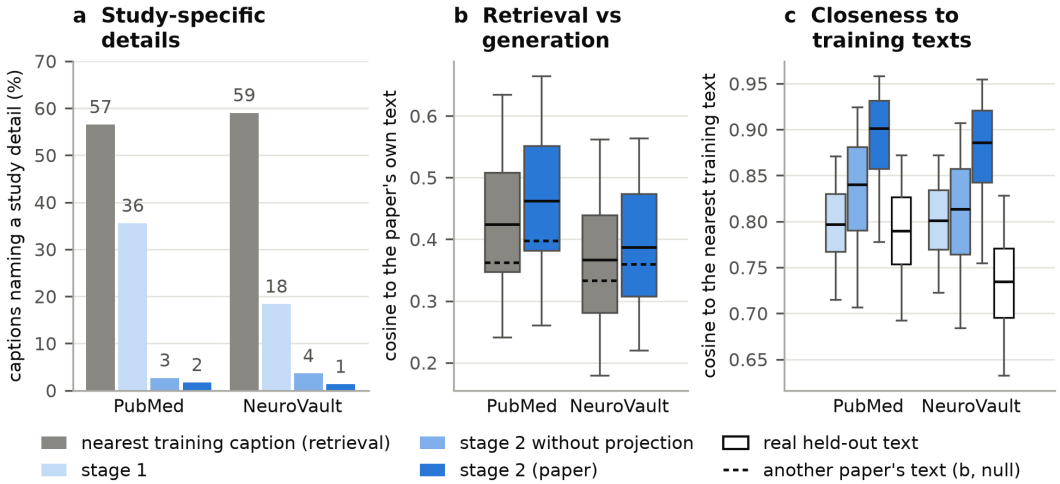


Fig. R13. Granularity of generated captions. Captions of PubMed test maps and NeuroVault papers, retrieved or generated at each stage. Panels show (a) the share naming a study detail and the MiniLM cosine (b) with the paper's own text and (c) with the nearest training text.

Table R11. Generated versus retrieved captions. Texts returned for PubMed test maps, scored against each paper's title and LLM summary by MiniLM cosine and BERTScore F1. Brackets give the MiniLM cosine against another paper's text.

Text returned for a test map	MiniLM cosine (null)	BERTScore F1
NeuroVLM, stage 2 caption (paper)	0.463 (0.397)	0.553
NeuroVLM, stage 1 caption	0.431 (0.358)	0.577
Retrieval, training caption nearest to the map	0.428 (0.362)	0.588
Random training text	0.363	0.580

Changes to the manuscript: Methods, Text Decoder, l. 316–320, on the canonical projection, which removes study details but adds near-copies. Discussion, l. 863–868. New Supplementary Section Caption Granularity, l. 2106–2115, with Fig. S10 (p. 47).

3.4: Context-free baseline

Reviewer's comment

Another capability that sets NeuroVLM apart is the context and flexibility provided by the LLM encoder. However it would be helpful to directly quantify how the LLM encoder performs relative to context-free baselines (such as mean Word2Vec embeddings) on the retrieval and text-to-neuroimage tasks.

Response: SPECTER2 outperforms a context-free baseline on both tasks, by a modest margin (Fig. R14). We replaced SPECTER2 with the mean word2vec embedding of each title and abstract and retrained only the projection heads. Mean retrieval AUC, the area under the recall curve, is 0.83 against 0.81. Text-to-brain Dice and voxel correlation are also higher with SPECTER2. We added this comparison to the Discussion and the Supplementary Information.

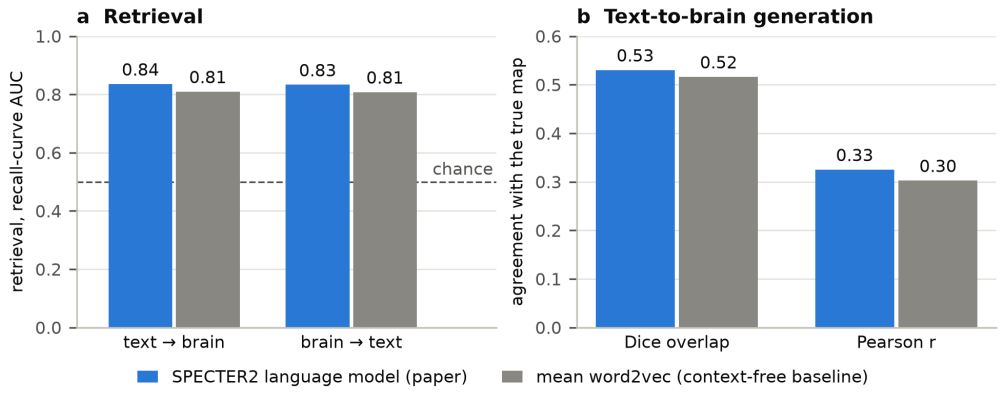


Fig. R14. Context-free encoder. PubMed test-set performance with SPECTER2 or with the mean word2vec embedding of each title and abstract, after retraining only the projection heads. Panels show (a) retrieval AUC in both directions and (b) text-to-brain Dice and voxel correlation.

Changes to the manuscript: Introduction, l. 109–111, and Discussion, l. 850–856. New Supplementary Section Context-Free Encoder, l. 2145–2151, with Fig. S11 (p. 48).

3.5: Interpretation, perturbations, and ablations

Reviewer's comment

It would be helpful to include more analyses like Figure 6 that interpret different components of the model. Interpreting the encoder weights could reveal the specific associations between text and neuroimage elements that NeuroVLM learned. Perturbing the inputs (such as adding negations) could show how NeuroVLM goes beyond linear additivity. Providing ablation results in the supplementary materials could compare the importance of different modeling decisions (such as the cross-entropy loss and canonical projection matrix).

Response: 1. Encoder weights. We interpret the latent dimensions that texts and maps share, rather than raw weights. For the clearest dimensions, the decoded map matches the Neurosynth map of the dimension's top term, such as listening in superior temporal cortex (Fig. R15, $r = 0.65$). Text and maps agree on what each dimension means. We added this figure to the Supplementary Information.

2. Perturbations. NeuroVLM responds to word swaps and to joined concepts (Fig. R16) but ignores negation. Swapping one word moves the map toward the matching Neurosynth contrast, and joining two concepts covers both. A sentence and its negation give nearly the same map ($r = 0.84$ to 1.00). We find no evidence of composition beyond addition. We added this limitation to the Discussion.

3. Ablations. The cross-entropy loss matters most for retrieval. Without it, retrieval AUC drops from 0.83 to 0.62, while the temperature and a one-directional loss barely matter (Table R12). The final text-to-brain head also beats heads trained with other losses. Comment 3.3 gives the effect of

the canonical projection on captions. We added these ablations to the Supplementary Information.

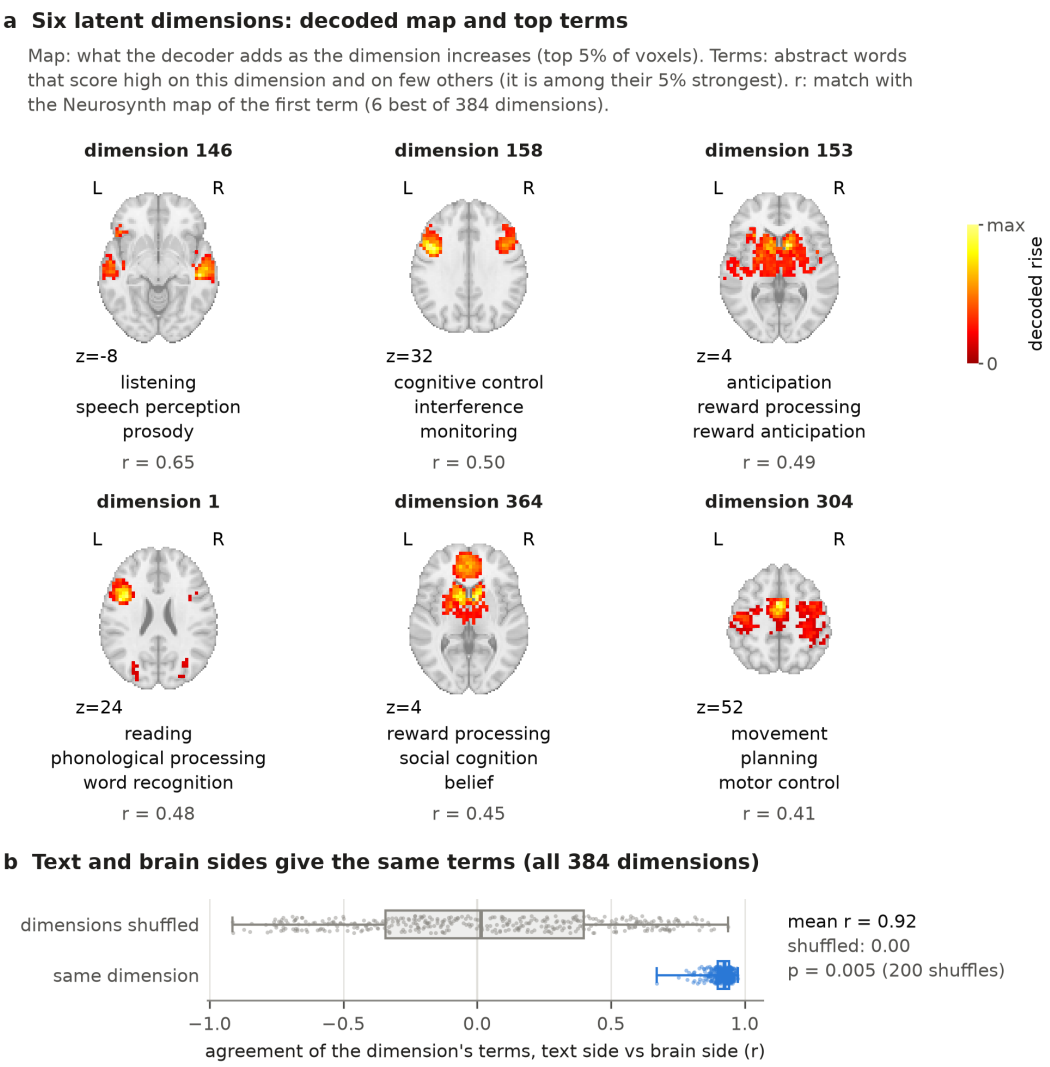


Fig. R15. Latent dimensions. (a) Decoded maps of six latent dimensions, each labeled with its three most associated Cognitive Atlas terms and its correlation r with the top term's Neurosynth map. (b) Agreement of each dimension's term associations between the text and brain sides, against shuffled dimensions.

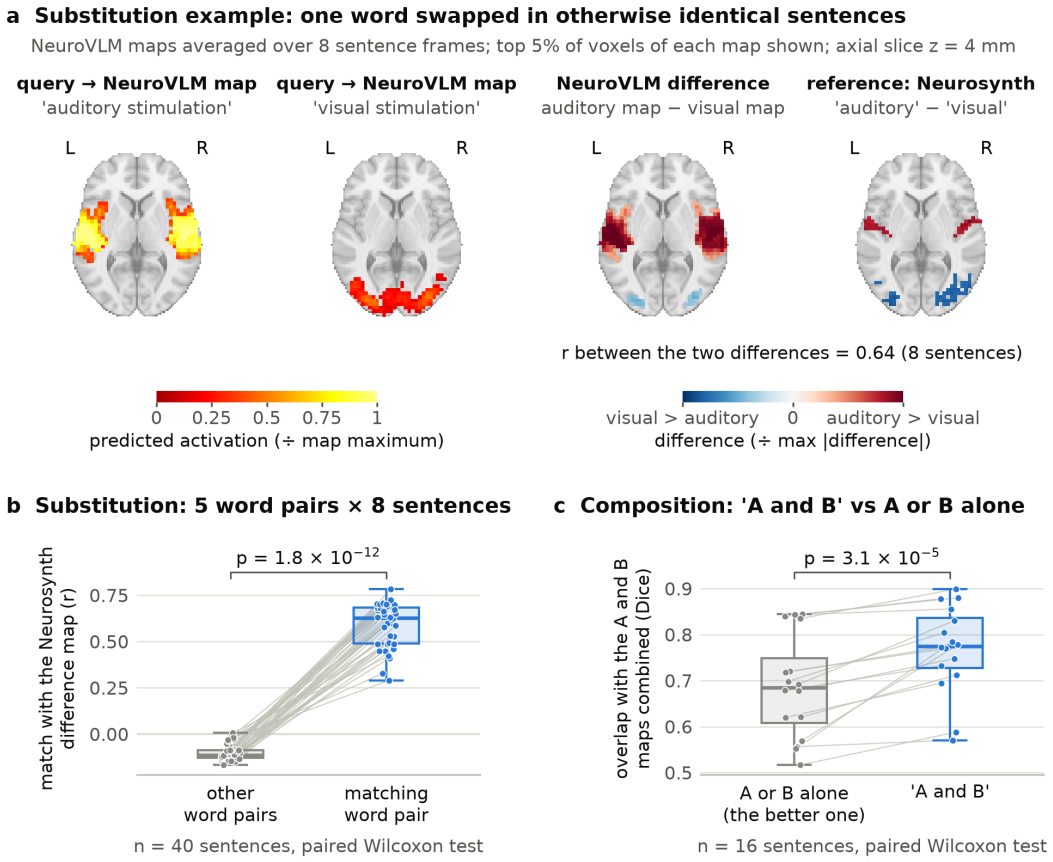


Fig. R16. Perturbation probes. (a) One word swapped, with NeuroVLM's difference map beside the matching Neurosynth contrast. (b) Match of such differences with Neurosynth for five word pairs, and (c) overlap of "A and B" maps with both concepts, against single-concept maps.

Table R12. Loss ablations. Heads retrained on the final model's latents with the loss shown and scored on the PubMed test set. Values are means of three seeds.

Loss	Test-set score
Contrastive heads, retrieval AUC	
Symmetric InfoNCE cross-entropy, $\tau = 0.07$ (final)	0.833
One direction only, text → brain / brain → text	0.834 / 0.827
$\tau = 0.03$ / $\tau = 0.20$ / learnable τ	0.829 / 0.827 / 0.833
No cross-entropy, cosine of matched pairs only	0.618
Text-to-brain head, Dice	
Latent MSE and map correlation, final head with its own training settings	0.530
Latent MSE only	0.512
Voxel-wise binary cross-entropy through the decoder	0.491
Latent cosine only	0.385

Changes to the manuscript: Introduction, l. 76–79, where the contrast with additive term models no longer implies non-linear composition. Discussion, limitations, l. 904–908, with a new sentence that NeuroVLM ignores negation. New Supplementary Section Latent Dimensions and Loss Ablations, l. 2186–2195, with Fig. S12 (p. 49) and Table S10 (p. 50).

3.6: Network, anatomy, and cognition outputs

Reviewer's comment

It is interesting that the model can generate separate brain-to-text outputs corresponding to the predicted network, anatomy, and cognition. How does quantitative performance compare between these outputs? What if the text-to-brain outputs were generated using only text about the network, anatomy, or cognition? These analyses would also shed light on the associations learned by NeuroVLM.

Response: We scored each generated title against its map. Every title is generated by the language model, steered toward the canonical vocabulary. The canonical projection steers the model's input, and at [NETWORK] and [REGION] titles, tokens that cannot continue a canonical name are masked. The model's own probabilities choose the name, so nothing is classified or looked up (2.12).

1. Comparison of the three outputs. [REGION] and [FUNCTION] titles are correct most often and [NETWORK] titles least (Fig. R17). A title is correct if it names the map's network, a region inside the map, or a function decoded from the map. On PubMed test maps, 63% of [REGION] titles are correct, against 24% by chance. Fig. R17 also scores atlas networks and ICA components.

2. Network, anatomy, or cognition text only. Maps generated from a network name, a region name, or a cognitive term alone all match their references better than a null baseline (Table R13). Region maps overlap their references least, because most regions are smaller than the top 10% of voxels that Dice compares.

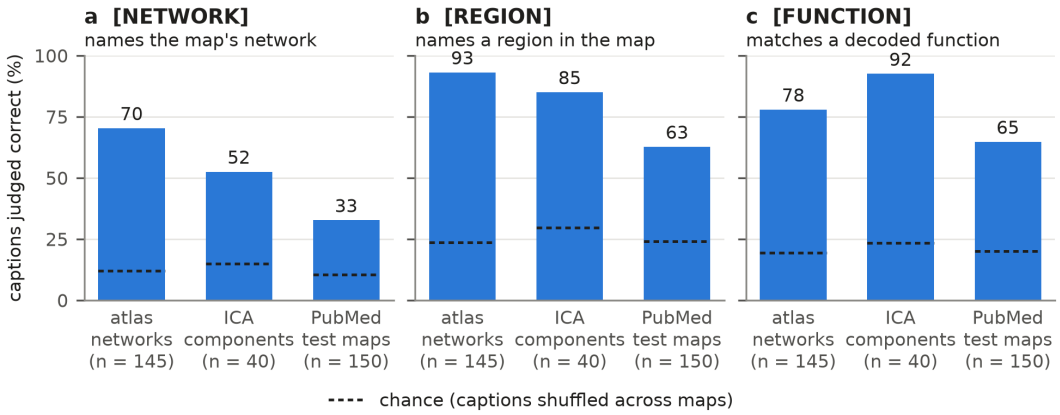


Fig. R17. Correctness of generated titles. Share of [NETWORK], [REGION], and [FUNCTION] titles judged correct for atlas networks, ICA components, and PubMed test maps. Dashed lines show chance, from titles shuffled across maps.

Table R13. Text-to-brain from one kind of text. Maps generated from bare network names, region names, or Cognitive Atlas terms, scored against the matching atlas map, region, or Neurosynth term map. Brackets give a null baseline, the same maps scored against the other items' references.

Input text	Reference	n	Dice (null)	Pearson r (null)
Network name	Yeo-7 and network atlas maps	126	0.29 (0.05)	0.29 (−0.01)
Region name	Harvard-Oxford and AAL regions	178	0.07 (0.01)	0.15 (0.00)
Cognitive Atlas term	Neurosynth map of the term	130	0.28 (0.13)	0.25 (0.02)

Changes to the manuscript: Methods, Text Decoder, l. 299–306, on title decoding steered toward the

canonical names. Discussion, limitations, l. 899–905, where [NETWORK] titles are the least reliable, especially for single-study maps.

3.7: Unsupported claims

Reviewer's comment

Some claims are not justified in the context of the study. For example "each component of NeuroVLM is modular and interchangeable with larger or smaller models depending on requirements" (line 101) and "our results underscore that the limiting factor for biological fidelity is often the structure of what the field reports and shares" (line 672). While intuitively true these statements would be better supported by including scaling and ablation analyses.

Response: 1. Modular and interchangeable components. The encoder swap of 3.4 supports a narrower claim, which we keep. Replacing SPECTER2 with word2vec required retraining only the small projection heads, and performance dropped modestly (Fig. R14). We softened the sentence to this claim and removed the reference to larger or smaller models, since we did not test scaling.

2. Limiting factor. The sentence compared coordinate-based supervision with full statistical maps, not model sizes. We rewrote it to say so and cited work on what coordinate-based meta-analysis loses.

Changes to the manuscript: Introduction, l. 109–111, modularity sentence. Discussion, l. 937–945, limiting-factor sentence, and l. 912–913, where the sentence on our strongest limitations is deleted.

3.8: Titles and abstracts only

Reviewer's comment

Could the authors justify why NeuroVLM was only trained on titles and abstracts? I imagine that other sources of information (such as table captions) might be useful for checking whether the activation coordinates are actually related to the claims in the titles and abstracts.

Response: Titles and abstracts exist for every paper, but table captions only for papers with open-access full text, fewer than half. Captions also say little about where coordinates lie, which limits their use for checking the abstract's claims (Table R14). A classifier predicts a table's main region from its caption at chance level (macro-F1 0.06 against 0.05), but does better from the abstract. We added these reasons to the Discussion.

Table R14. Uninformative table captions. Captions of coordinate tables from open-access papers, picked by hand from 60 random examples. The right column lists the Harvard-Oxford regions that hold most of each table's peaks.

Kind	Caption	Where its coordinates lie
boilerplate	Brain regions revealed by contrasts of interest.	anterior cingulate, frontal pole, insula
boilerplate	The main results of group analysis of block data.	posterior cingulate, amygdala, thalamus
boilerplate	Regional imaging relationships.	lateral occipital cortex, precuneus
condition code	Simple Effects contrast JUDGE –BODY.	anterior cingulate, lateral occipital, postcentral
condition code	Neural activations for the contrast AU > OC.	lateral occipital, paracingulate, posterior cingulate
statistics only	Functional results (p = .05 family-wise error (FWE) cluster corrected).	lateral occipital, pallidum
statistics only	Significant clusters exhibiting interactions of interest at $p < 0.001$ (uncorrected), n = 15.	insula, paracingulate, orbitofrontal
region list only	Significant clusters from amygdala connectivity analyses.	medial frontal cortex, insula

Changes to the manuscript: Discussion, l. 877–884, on table captions and on the pooling of a paper's tables as a limitation.

3.9: Percentiles of the qualitative examples

Reviewer's comment

It would be helpful to know where the qualitative examples in Figures 2 and 3 fall within the distribution of cosine similarity quality scores. Alternatively the authors could provide a percentile for each example.

Response: Table R15 gives each example's percentile within its set, from the 16th to the 99th. Fig. 3 also plots both similarity distributions and marks each example. Its rows follow a fixed rule. Each third of the distribution contributes the papers nearest its center with a clean generated text. High and Medium rows also require a title that names one domain and a true map with distinct clusters.

Table R15. Percentiles of the examples. Percentile of each example of Fig. 3 and Fig. 2 among all items of its set, by text-to-brain and brain-to-text cosine similarity. Fig. 3 ranks 305 NeuroVault papers and Fig. 2 ranks atlas networks.

Example	Text → brain percentile	Brain → text percentile
Fig. 3, High, working memory	86	79
Fig. 3, High, language	77	87
Fig. 3, Medium, memory	52	62
Fig. 3, Medium, reward	36	43
Fig. 3, Low, poor input map	16	18
Fig. 3, Low, model miss	22	19
Fig. 2, Visual	76	99
Fig. 2, Language	75	37
Fig. 2, Auditory	31	19
Fig. 2, Somatomotor	39	72
Fig. 2, Interoception	44	55
Fig. 2, Default mode	69	94
Fig. 2, Frontoparietal control	48	53
Fig. 2, Salience	33	46

Changes to the manuscript: Fig. 3, on one page, with rows chosen by a fixed rule and distribution strips that mark each example (p. 12). Fig. S4 caption, which gives the percentiles of its components (p. 34). Results, Generative Models: Networks, l. 445–447, with the percentiles of the Fig. 2 examples, and NeuroVault, l. 457–459, on the complete captions and the choice of rows.

3.10: Code availability

Reviewer's comment

I was able to install and run the code. The project is very well-documented and user-friendly. I believe that it provides a very usable resource for the community.

Response: No change was requested. The reproducibility changes of comment 1.11 also apply here.